

Numbers You Should Know

Part 1

A semi-quantitative profile of the profession

Sources?

- American Astronomical Society
- American Institute of Physics
- NSF, NASA, other agencies
- UVa documents
- Literature and general media sources
- ROMEs = rough order magnitude estimates

PRESENTATIONS AND RESOURCES

A. Introduction

[Course Assignments](#)

[Optional Topic List](#)

B. Profile of the Profession

[Numbers You Should Know](#) (O'Connell 2026)

Budget Outlooks 2022-2024:

[NSF Outlook](#) (AAS Town Hall, Jan 2022)

[NSF Astronomy Budget Outlook](#) (Sep 2023)

[NASA Outlook](#) (AAS Town Hall, Jan 2024)

[Astronomy NSF/NASA Budget Outlook](#) (Sep 2024)

Literature on Careers in Astronomy:

["Production Rate and Employment of PhD Astronomers"](#) (Metcalf, PASP, 120, 229, 2008)

["A Closer Look at Astronomy Faculty"](#) (Ivie, AIP, 2009)

[BA Degree Gender Gap by Field, 1971-2017 \(graphic\)](#) (Perry, 2019)

["Astronomy Enrollments and Degrees"](#) (Mulvey & Nicolson, AIP, 2014)

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["Women's and Men's Career Choices in Astronomy and Astrophysics"](#) (Ivie et al., PRPER, 12, 020109, 2016)

["A Survey of U.S. Astronomers"](#) (Spuck, PhD Thesis, WVU, 2017)

["Degree Plus One, Employment & Salaries"](#) (Mulvey & Pold, AIP, 2019)

["Long Term Trends in the Astronomical Workforce"](#) (Momcheva, Astro 2020 White Paper)

["Gender & the Career Outcomes of PhD Astronomers in the US"](#) (Perley, PASP, 131:114502, 2019)

[US Astronomy PhD Employment \(graphic\)](#) (Ivie & Pold, AIP, 2021)

["Workforce Summary of 2021 US AAS Members, Summary Results"](#) (Ivie & Pold, AIP, 2021)

["New Astronomy PhDs: What Comes Next"](#) (Mulvey & Pold, AIP, 2023)

Teaching Non-Majors Undergraduate Courses:

["Facts of Life for New Teachers in the Astronomy Nonmajors Curriculum"](#) (O'Connell, AstEdRev, 6, 1, 2007)

[Lecture version](#) (2025)

Science Outlook 2026+:

[News Clips on Trump Administration Science Policy](#)

[A Good Time to Start a PhD](#) (Palmer, 2025)



A National Perspective on Astronomy



21

Century 21

An apex of astronomical discovery!

Cosmology, galaxy formation, galaxy evolution,
black holes & neutron stars, astrochemistry,
planet formation, exoplanets, exobiology,
gravitational waves, SS planetary exploration

We're #3!

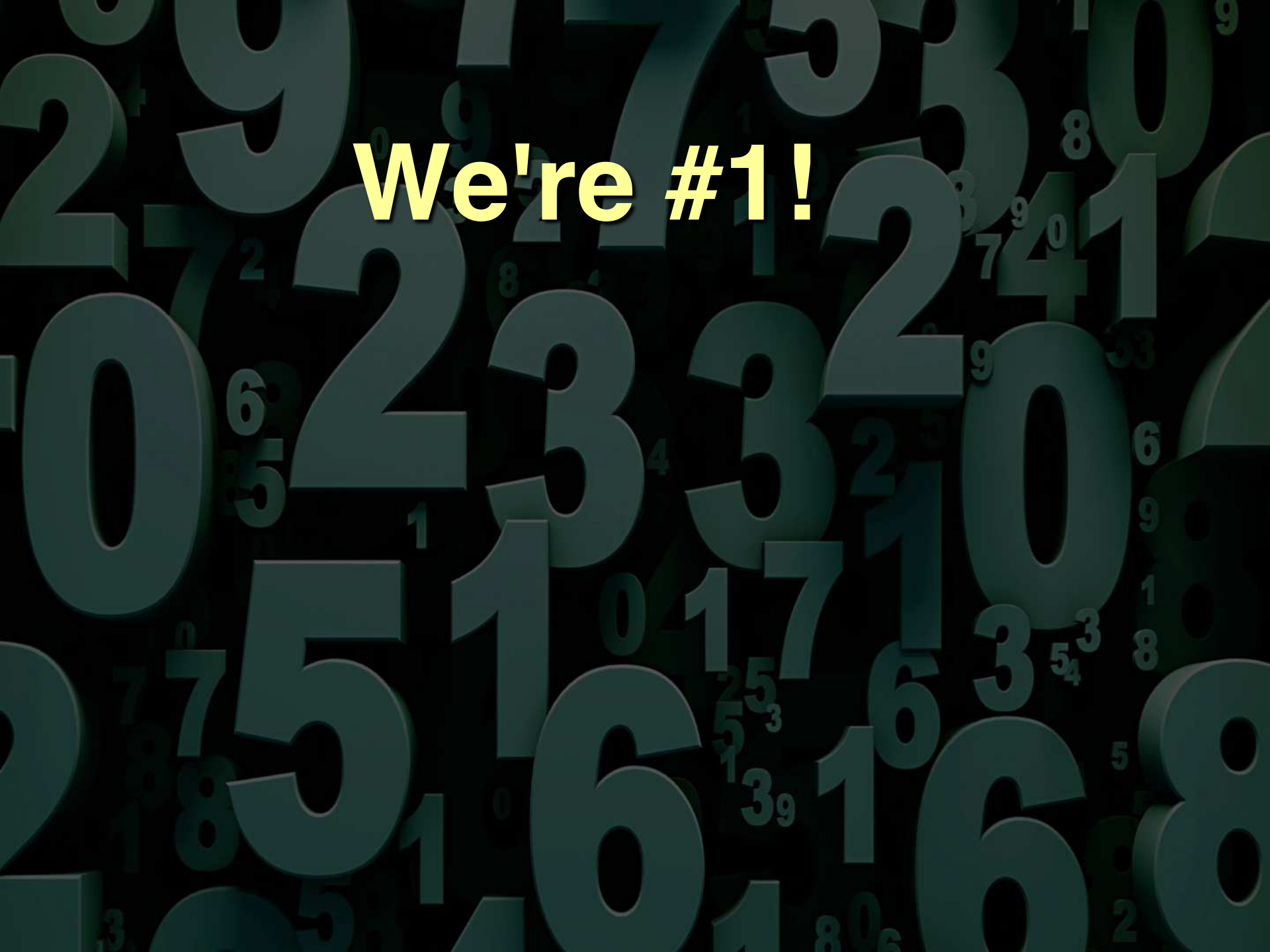
We're #3 among STEM fields in media impact:

#1 Health & Medicine

#2 Environment

#3 Astronomy & Space

We're #1!

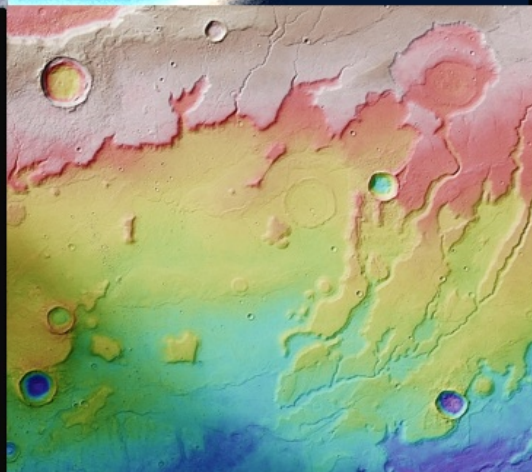
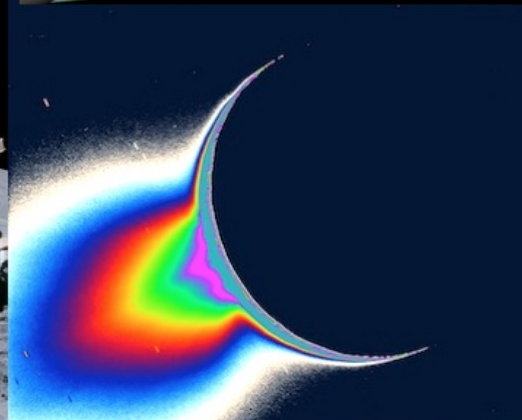
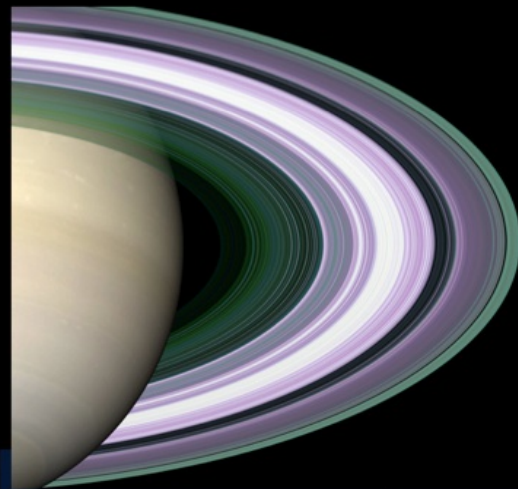
The background of the image is a dark, textured surface covered with numerous 3D-rendered numbers in various sizes and orientations. The numbers are in shades of dark green and black, creating a sense of depth and complexity. The text 'We're #1!' is prominently displayed in the upper center in a bright yellow, bold, sans-serif font.

**We're #1 in impact
per practitioner!**

Total STEMM employment: 16 million

Total astronomy & space science
employment: ~20-30 thousand





SUN 9/8c MAR 9

COSMOS: A SPACETIME ODYSSEY

Presented by FOX Sun 9/8c and National Geographic Mon 10/9c

[Blog](#) [Clips](#) [Live Event](#)



Samsung GALAXY

Jeep

AN EPIC
ADVENTURE IN TIME,
SPACE AND LIFE.

C

O

S

M

O

S

A SPACETIME ODYSSEY

SUNDAY MARCH 9 ON 10 CHANNELS

FOX

NATIONAL
GEOGRAPHIC
CHANNEL

NAT GEO
WILD

NAT GEO
mundo

FOX life

FX FXM

FOX
7

FOX
2



Eclipse USA 2017



215 Million Americans Watched the Solar Eclipse, Study Finds



Todd Heisler/The New York Times

By Jonah Engel Bromwich

Sept. 27, 2017



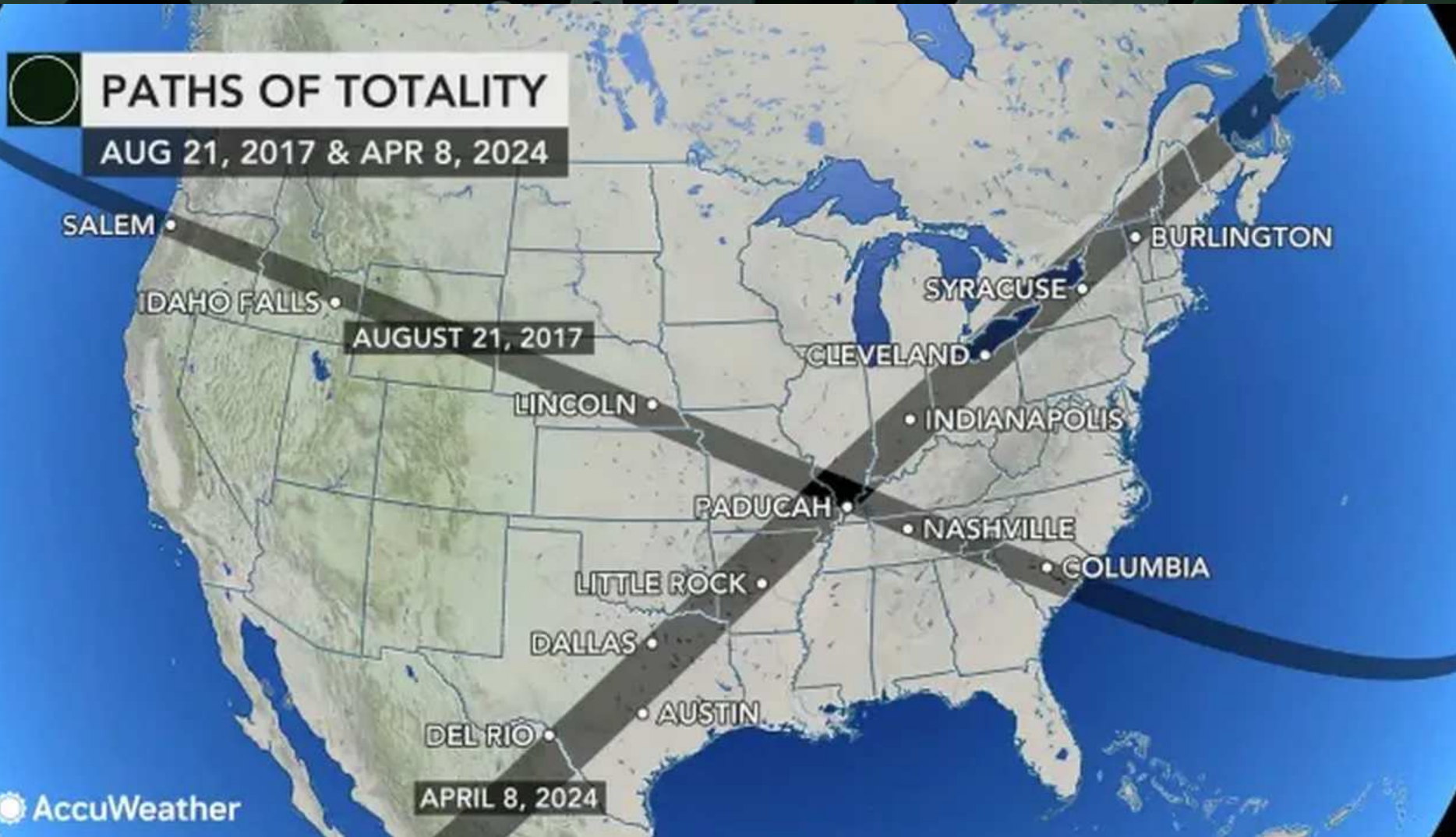
We hear it all the time: Americans are [more divided than ever](#), or at least since the Civil War.

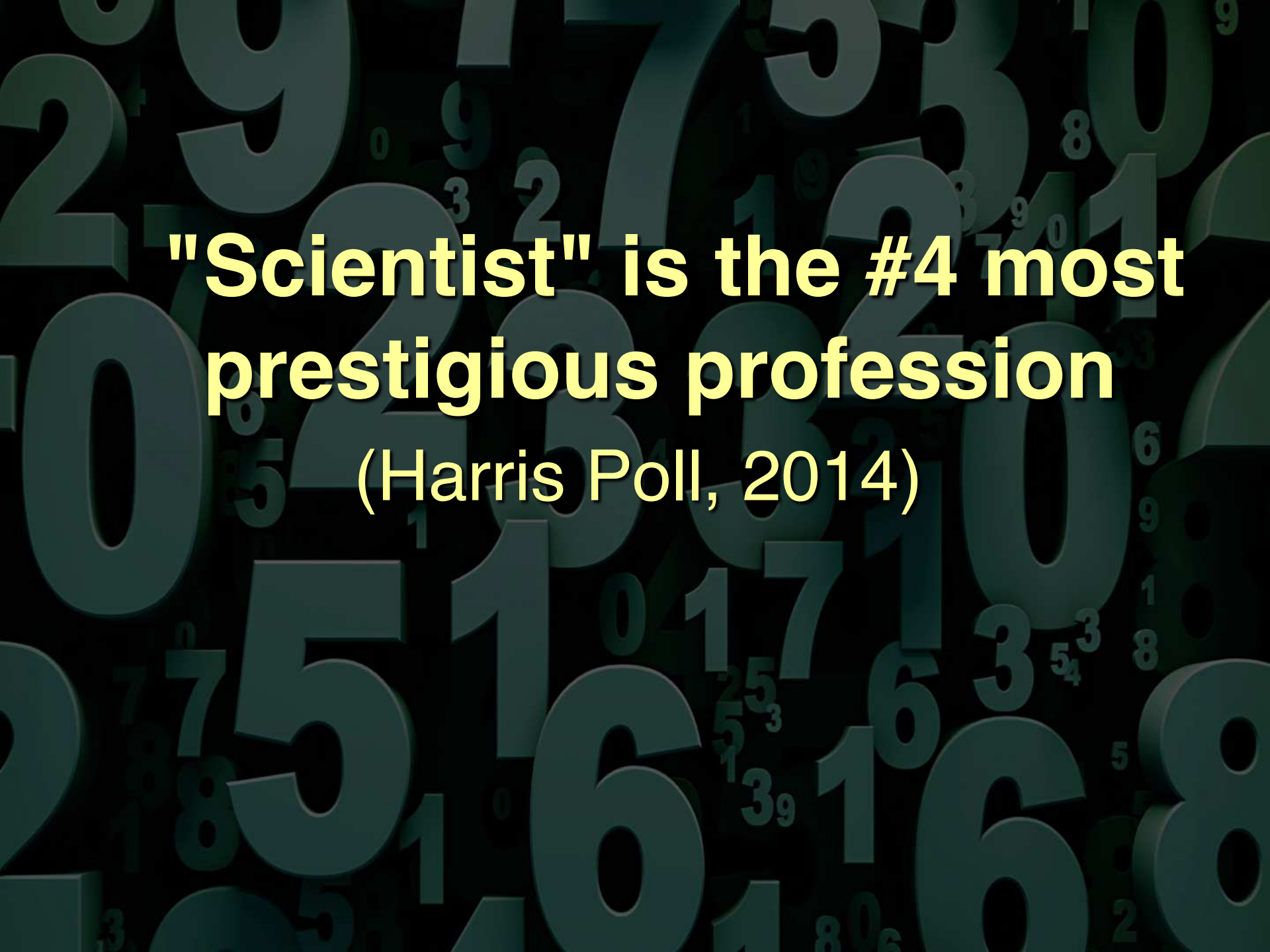
But [the solar eclipse on Aug. 21](#) brought the United States together in greater numbers than most any national event in recent memory, according to a study released Tuesday by the University of Michigan. It estimated that 88 percent of American adults — about 215 million people — watched the solar eclipse, either in person or electronically.

That's nearly twice the number of people that watched the Super Bowl last year. It's [almost 30 percent more Americans](#) than participated in the presidential election last year.



Did it again! April 2024



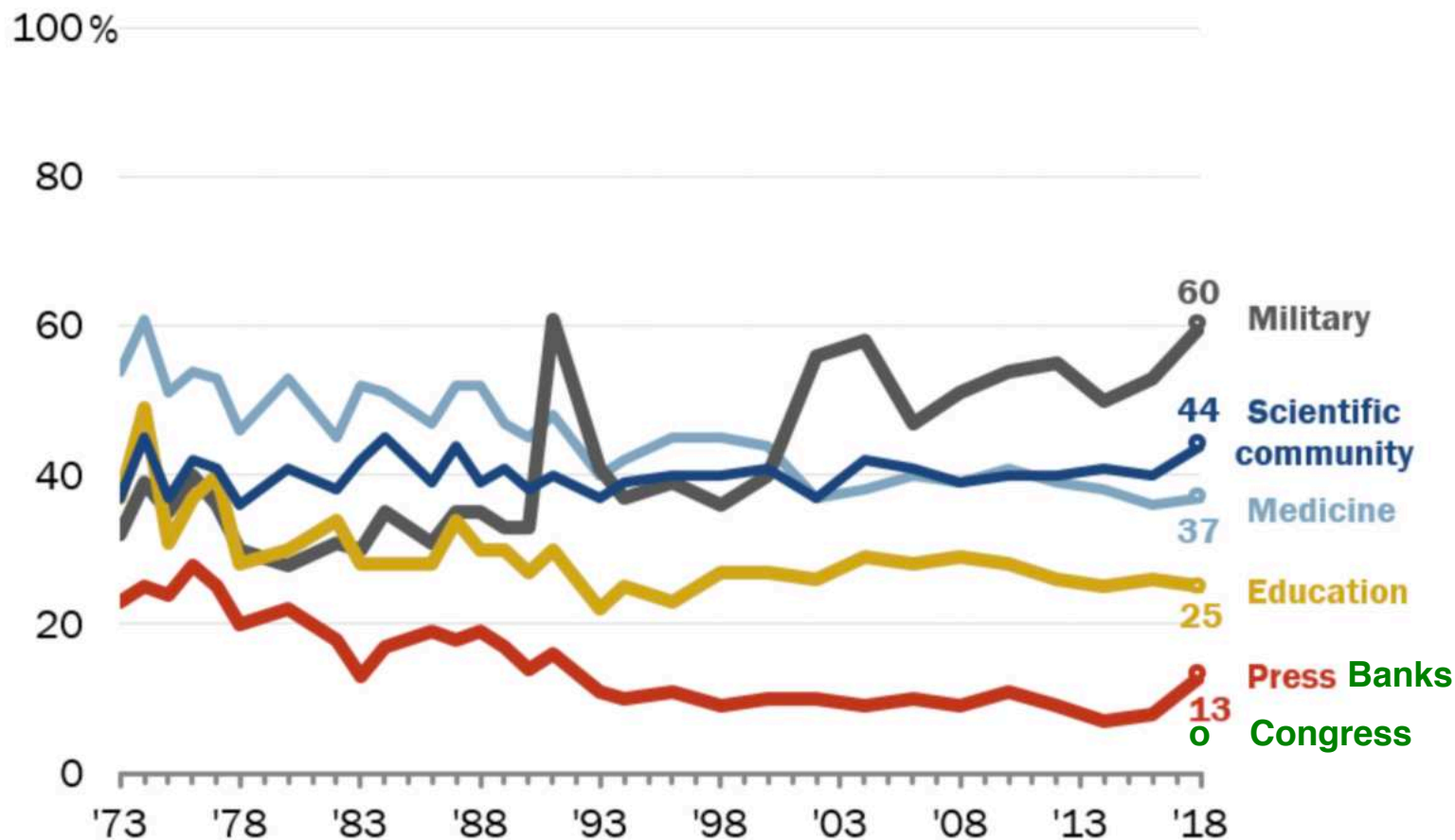
The background of the slide is a dark, textured surface covered with numerous 3D-rendered numbers in various shades of teal and green. The numbers are of different sizes and are scattered across the entire frame, creating a sense of depth and complexity. Some numbers are in the foreground, appearing larger and more detailed, while others are in the background, appearing smaller and more faded. The overall effect is a dense, abstract pattern of digits.

**"Scientist" is the #4 most
prestigious profession**

(Harris Poll, 2014)

Confidence in leaders of the military has gone up; confidence in some other institutions is declining

% of U.S. adults who say they have a great deal of confidence in the people running each of these institutions



Note: Respondents who gave other responses or who did not give an answer are not shown.
Source: General Social Surveys, NORC.

PEW RESEARCH CENTER (2018 data)

250,000

= Number of college students
enrolled annually in elementary
astronomy courses



\$425,000

= National budget for astronomy
research per astronomer

THE NATIONAL BUDGET FOR ASTRONOMY (2016)

NSF	NASA	DOE, DOD	Univ/Priv*	Total**	Number Astronomers***	\$/Astronomer
\$250M	\$2950M	~\$50M	~\$150M	\$3400M	~8000	\$425,000

*Research support; excludes basic faculty salaries.

**The federal budget for astronomy is ~0.08% of the total federal budget of \$4.0T or \$10.09 per US citizen per year.

***AAS membership, 2016



\$425,000

= National budget for astronomy
per astronomer

...BUT: mostly in the form of *shared
observing facilities*

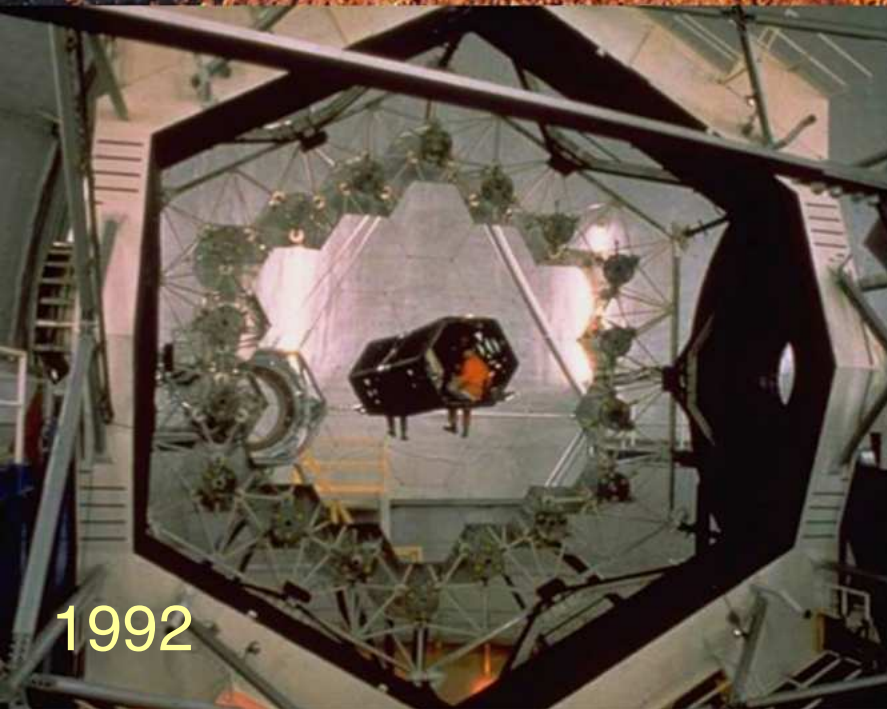
1974



1990



1992



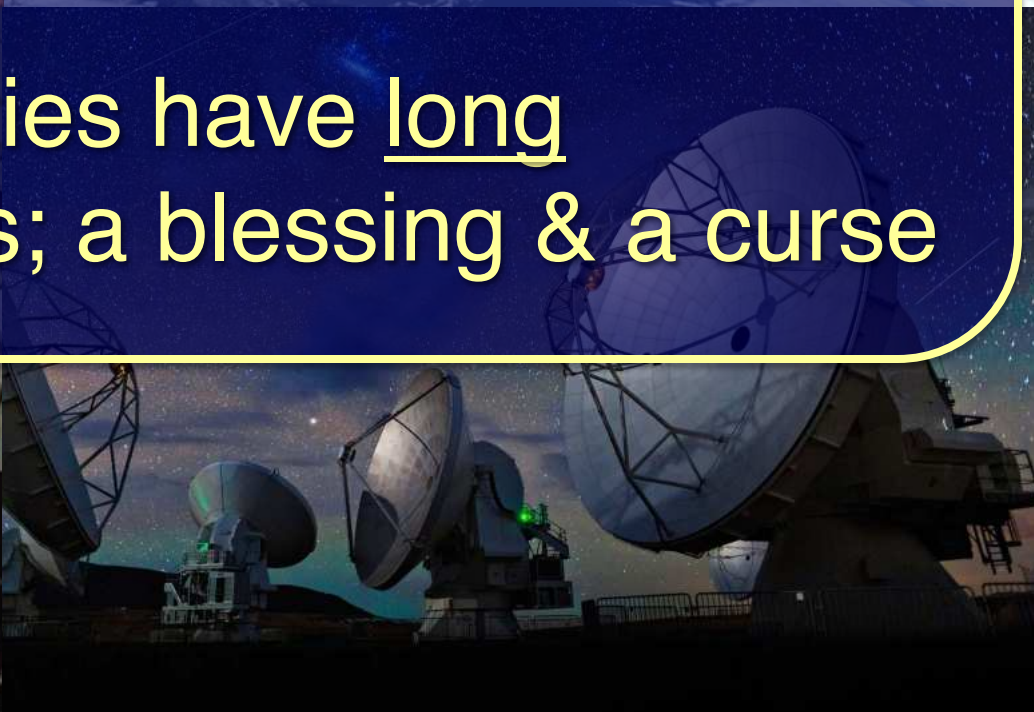
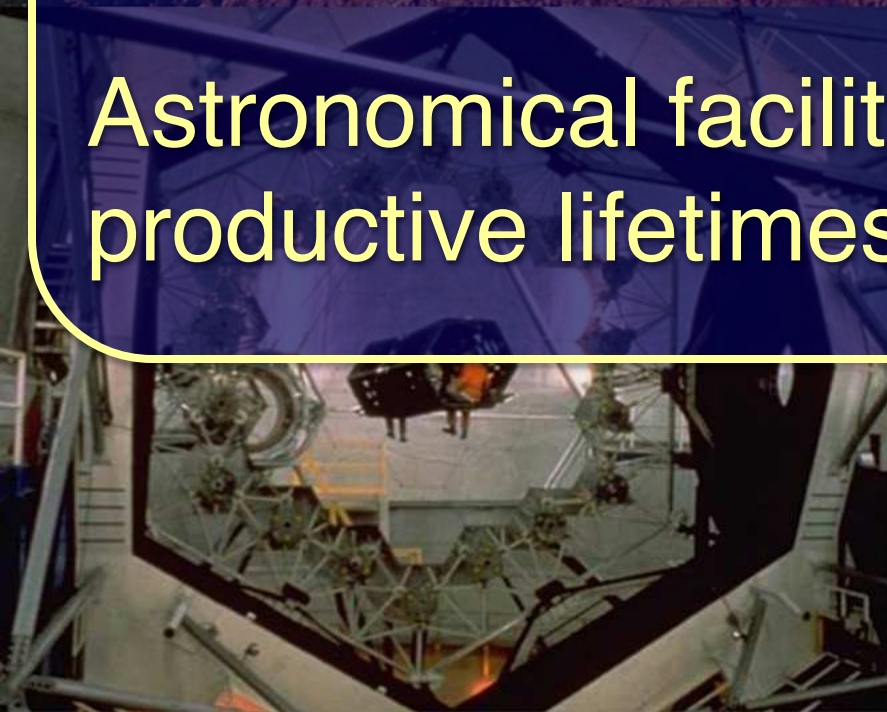
2011



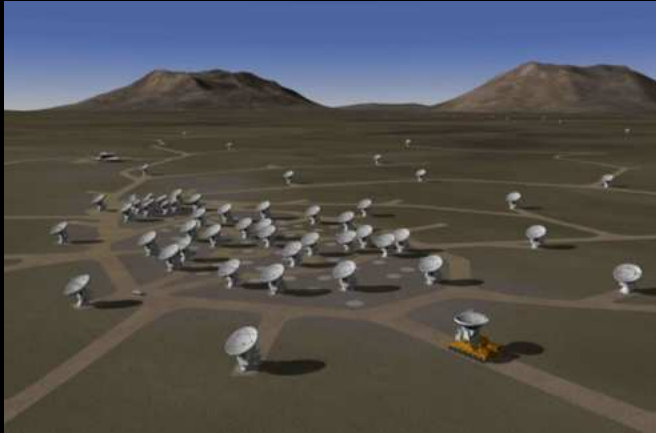


Astronomy is ~uniquely dependent on large, shared experimental facilities

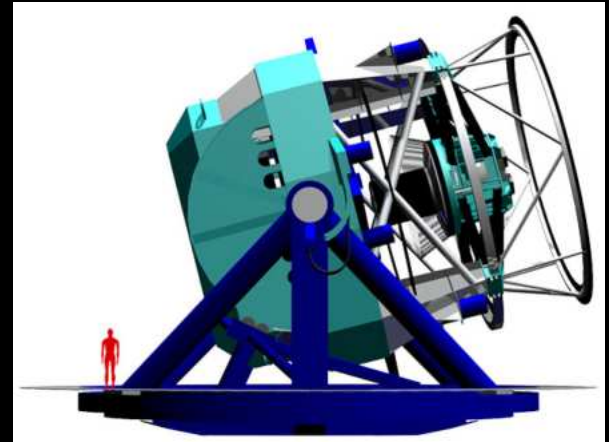
Astronomical facilities have long productive lifetimes; a blessing & a curse



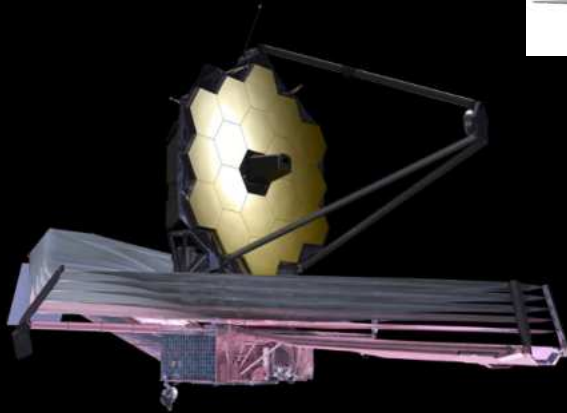
Promise of the 2020's



ALMA



Rubin/LSST



JWST



GMT



Roman/WFIRST

A high-angle, wide shot of a massive crowd of people, densely packed together. Many individuals are holding up small Union Jack flags, creating a sea of red, white, and blue. The crowd appears to be at a large public event or rally. In the center of the image, there is a white rectangular box containing the text "The Job Market" in a bold, blue, italicized font.

The Job Market

WHEW!



JAMES WEBB SPACE TELESCOPE

344 possible single point failures

....all avoided

\$60M/year for GO programs



JWST: SMACS 0723



JWST: M74 NIR-MIR

Kinds of Jobs for PhD Astronomers

- **Postdoctoral**
 - Short-term (1-3 yr) research positions (mostly directed)
- **Research Scientists**
 - Mostly semi-permanent. Large range, from support to independent researchers. Universities, observatories, government labs (e.g. NRAO, NOIRLab, GSFC, STScI, USNO). Independent contractors (e.g. JHU/APL, SWRI)
- **University Faculty**
 - Short term contractual and permanent (tenured)
Research and teaching
- **Non-astro-research Government**
 - E.g. NASA, NSF, DOD, DOE, NOAA, etc.
- **Non-astro-research Private Sector**
 - "Beltway Bandits," high-end computing, aerospace, sensors & optics, medical imaging, communications...

Kinds of \$\$\$

- **"Hard" money (reliable, long-term)**
 - Tenured faculty
 - Civil servants
 - Tenured & senior staff at national labs
- **"Soft" money (term-limited, grants, contracts)**
 - Postdocs
 - "Adjunct" faculty
 - Many "research scientists"
 - Federal contractors (e.g. SWRI)
 - Other private sector

2%

= Unemployment rate for astronomers

~ Transition rate

→ ~ Full employment

2/3

Astronomy Long-term PhD Employment Pattern Through the 1990's:

~1/3 Faculty

~1/3 Research Scientists

~1/3 Non Astronomy

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~1/3 Faculty

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} 65%

(Perley 2019)

Employment Statistics – Sources

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["Gender & the Career Outcomes of PhD Astronomers in the US"](#) (Perley, PASP, 131:114502, 2019)

C. Navigating the Early Career Job Market

[Preparing for and Navigating the Postdoc/Early Career Job Market \[encrypted\]](#) (Cleeves and Loomis)

[Applying for a Postdoc: Some Collected Experience](#) (Meyer and McGuire)

[AAS Job Register](#)

[Astronomy Rumor Mill](#) (astrobetter)

["The Professor Is In: The Essential Guide to Turning Your Ph.D. Into a Job"](#) (Kelsky)

D. Faculty "Top-10" Advice for Graduate Students

[Consolidated Faculty Advice, 2022](#)

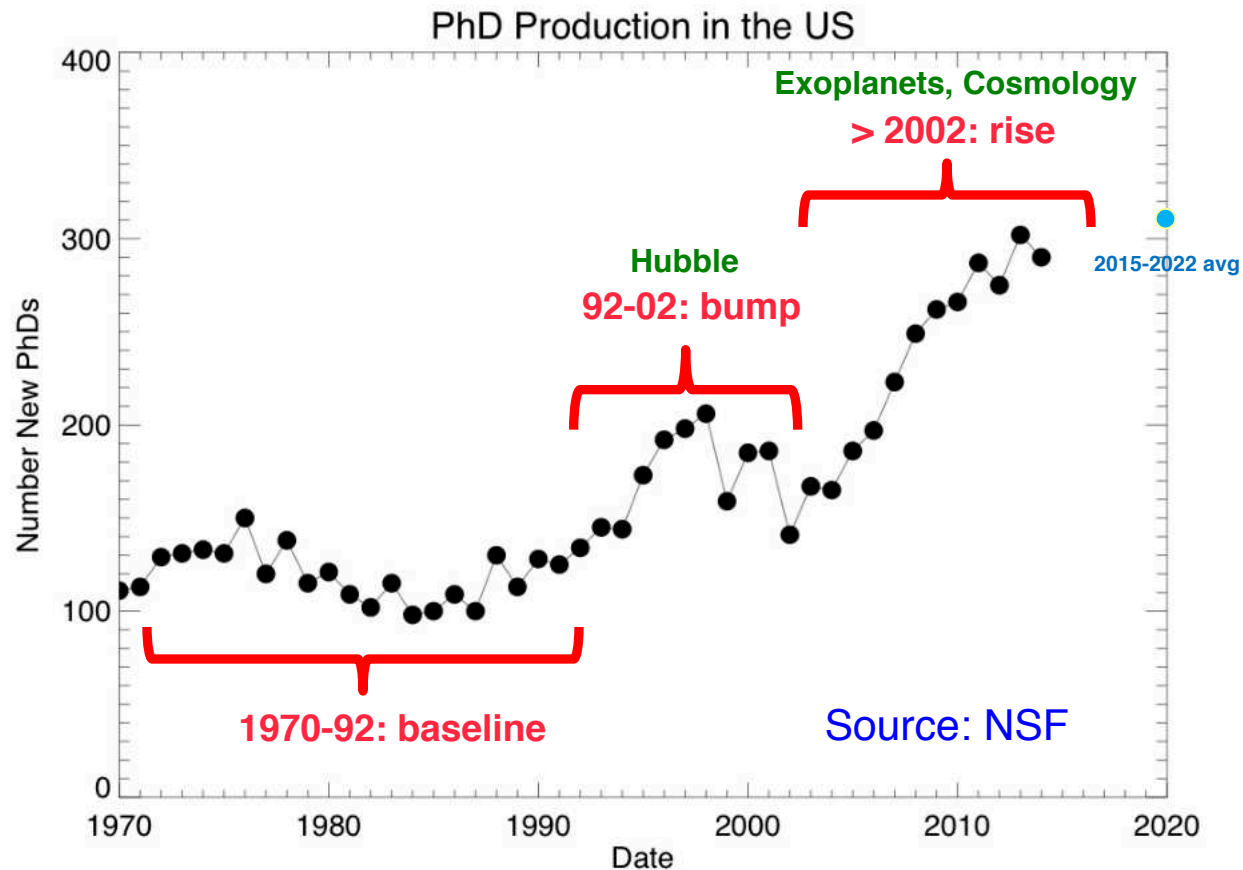
["Tips for Success in Observational Astronomy"](#) (O'Connell)

How many jobs?

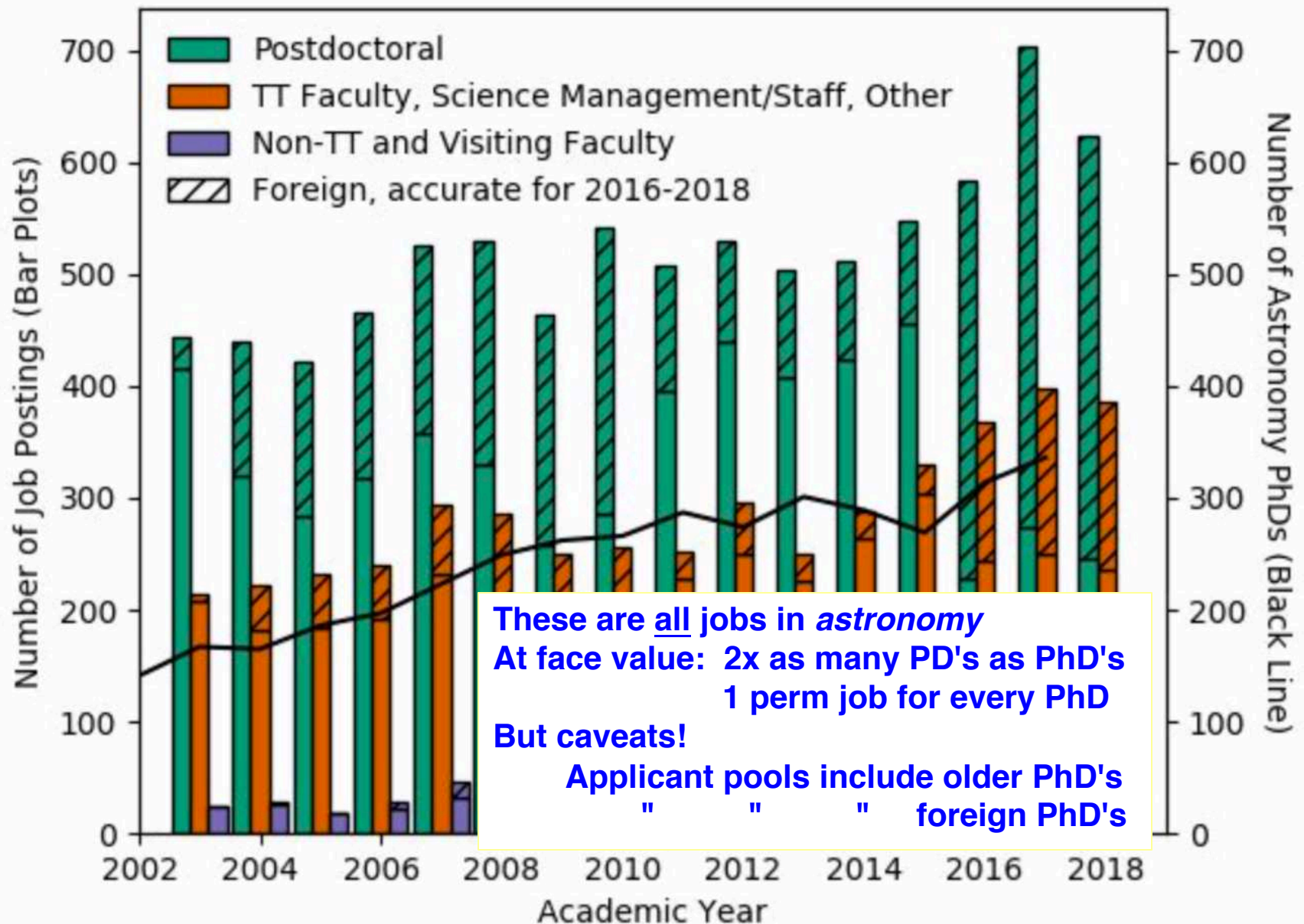
~8500 (US)

- = Membership of AAS + Nonmembers
- Non-grad degrees

Production of New Astronomy PhD's



AAS Job Register Statistics



Black line: PhD's produced

(Kamenetzky, White Paper, 2019)

Employment Demographics Studies

Perley 2019

Publications of the Astronomical Society of the Pacific, 131:114502 (7pp), 2019 November

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<https://doi.org/10.1088/1538-3873/ab0cc4>



Gender and the Career Outcomes of Ph.D. Astronomers in the United States

Daniel A. Perley 

Astrophysics Research Institute, Liverpool John Moores University, IC2, Liverpool Science Park, 146 Brownlow Hill, Liverpool L3 5RF, UK; d.a.perley@ljmu.ac.uk

Received 2019 February 10; accepted 2019 February 25; published 2019 September 24

Abstract

We analyze the postdoctoral career tracks of a nearly complete sample of astronomers from 28 United States graduate astronomy and astrophysics programs spanning 13 graduating years ($N = 1063$). A majority of both men and women (65% and 66%, respectively) find long-term employment in astronomy or closely related academic disciplines. We find no significant difference in the rates at which men and women are hired into these jobs following their Ph.D.s or in the rates at which they leave the field. Applying a two-outcome survival analysis model to the entire data set, we measure a relative academic hiring probability ratio for women versus men at a common year-post-Ph.D. of $H_{F/M} = 1.08^{+0.20}_{-0.17}$ and a leaving probability ratio of $L_{F/M} = 1.03^{+0.31}_{-0.24}$ (95% CI). These are both consistent with equal outcomes for both genders ($H_{F/M} = L_{F/M} = 1$) and rule out more than minor gender differences in hiring or in the decision to abandon an academic career. They suggest that despite discrimination and adversity, women scientists are successful at managing the transition between Ph.D., postdoctoral, and faculty/staff positions.

Key words: sociology of astronomy

Online material: color figures, machine-readable table

Employment Demographics Studies

Perley 2019

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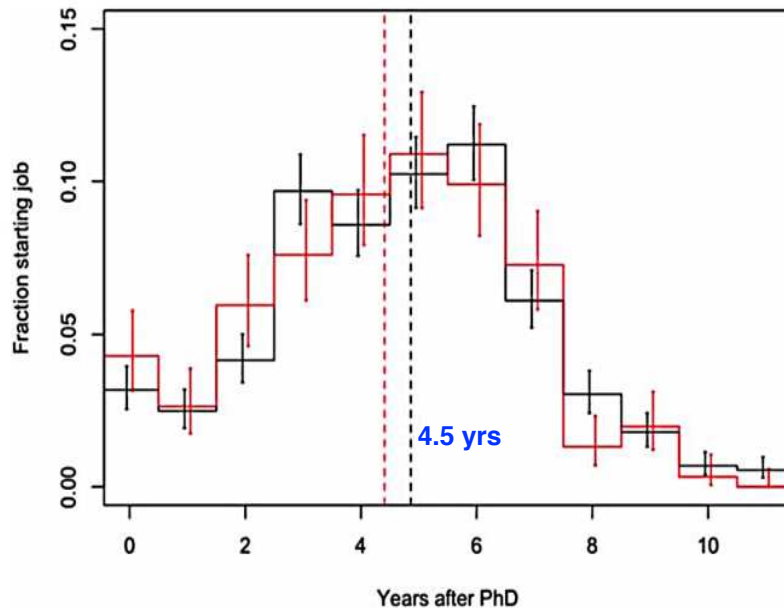
Followed actual post-PhD histories of over 1100 individual astronomers, unlike earlier studies using broad statistical measures or unreliable reporting (e.g. the "Rumor Mill" site).

Data for 2000-2012.

Perley 2019

12-Year Post-PhD Statistics (Data 2000-2012)

Frac starting perm job



Frac leaving field

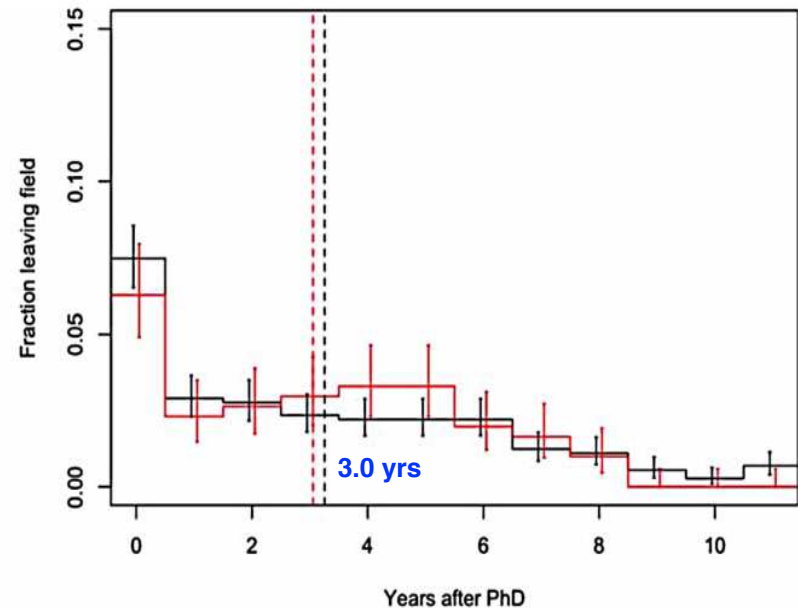
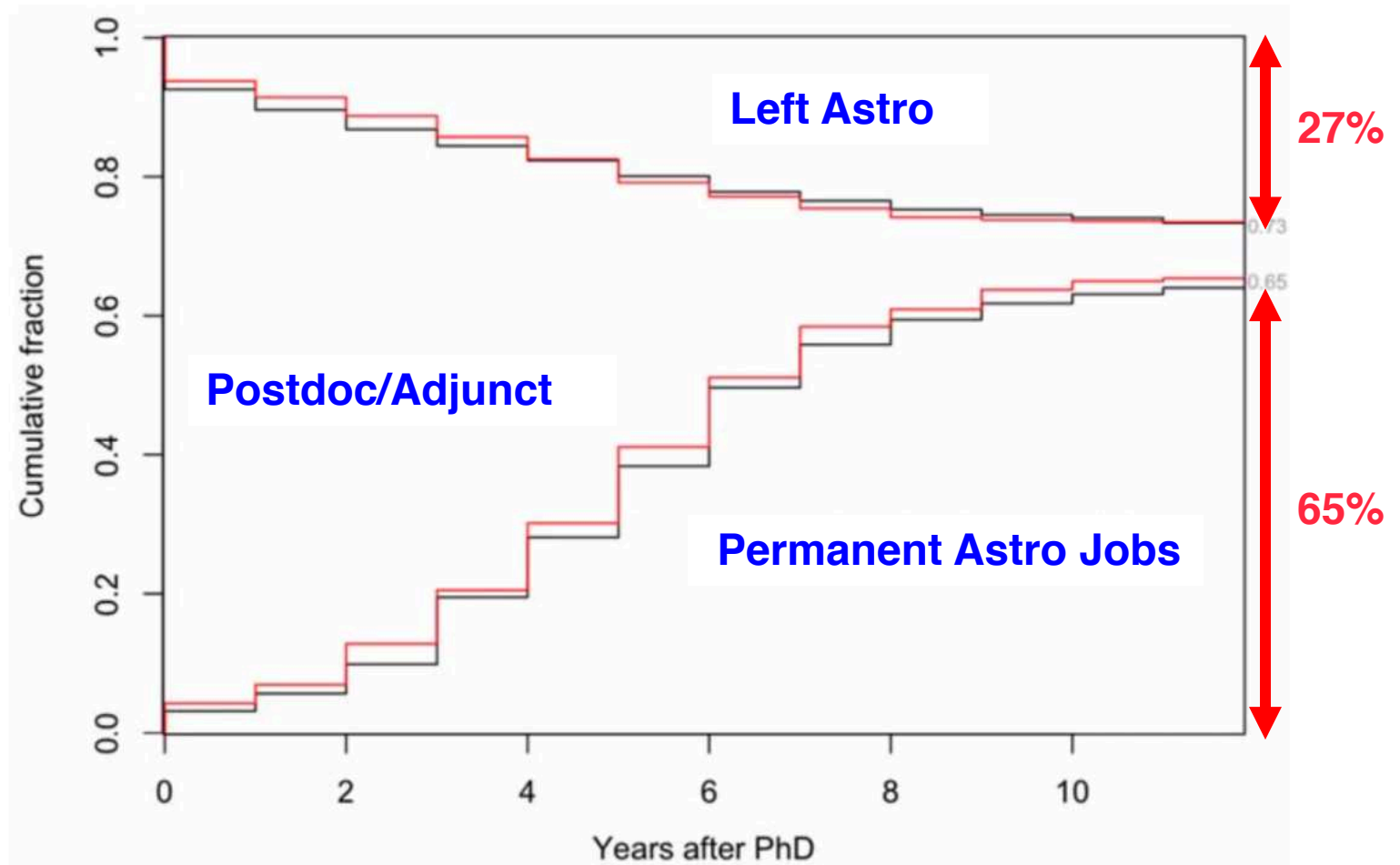


Figure 1. Histograms of recorded times (years after Ph.D.) at which Ph.D.s either: (left) progressed from term-limited to long-term or permanent positions within astronomy, or (right) left the field to pursue other employment. Histograms are normalized using total counts for each gender (regardless of outcome). Error bars show 67% binomial confidence intervals and dashed vertical lines show the means. Male astronomers are shown in black and female astronomers in red.

Perley 2019

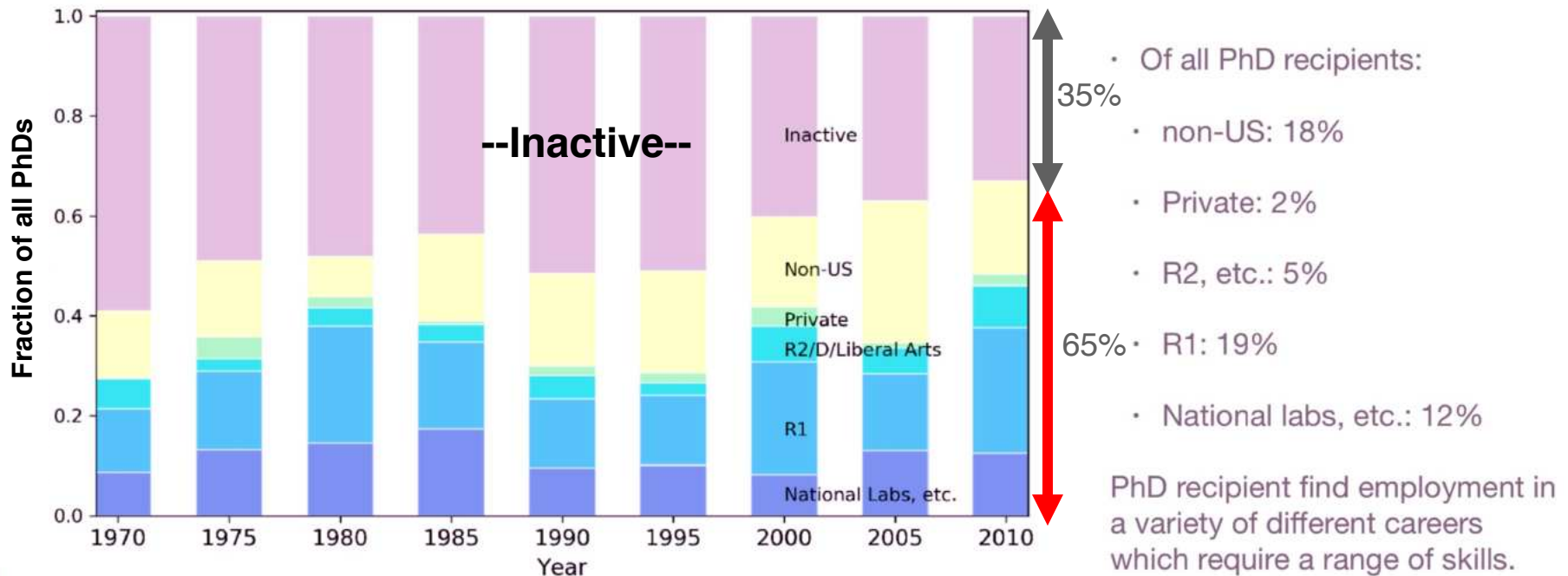
12-Year Cumulative Employment (Data 2000-2012)



Employment Demographics Studies

Momcheva 2019

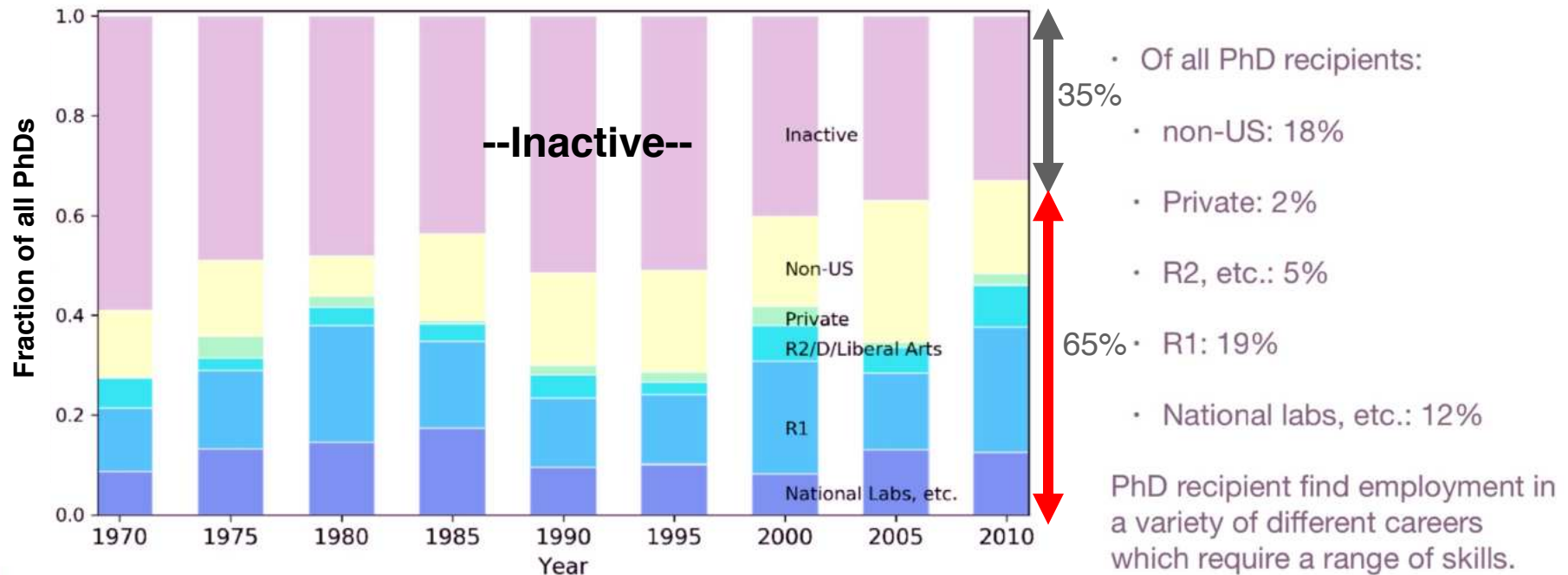
"Active" = published in professional literature within 3 yrs



Employment Demographics Studies

Momcheva 2019

"Active" = published in professional literature within 3 yrs



Result similar to Perley: 65% active

Perley (2019): *"The number of astronomy PhD's is not greatly in excess of the number of careers available within the field."* [Data for 2000-2012]

Kamenetzky (2019): *"The overall number of potentially permanent positions...has slightly increased in the past decade [2010-2019] to ~380 per year compared to ~270 ten years earlier, roughly keeping pace with the increase in new PhDs."*

permanent jobs in astronomy

Decadal Survey (2020): *"There is no evidence of mismatch between the number of PhD- or postdoc-trained astronomers and the broad array of desirable career pathways into the STEM workforce."*

Perley (2019): "The number of astronomy PhD's is not only in excess of the number of careers available within the field." [Data for 2000]

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permanent jobs in astronomy

Decadal Review: "There is no evidence of mismatch between the number of PhD- or postdoc-trained astronomers and the broad array of desirable career pathways into the STEM workforce."

**CONCLUDE: NOT GLOOMY, NOT FLUSH.
TEMPERED OPTIMISM**

Positions Held by UVa Astro PhD's (1967-2017; 130 Degrees)

Faculty	33%
Research scientists	40%
Non-Astro	13%
Outreach	2%
Secondary ed	1%
Postdocs	11%

Numbers You Should Know Part 2

Professional effort

ASTR 8500

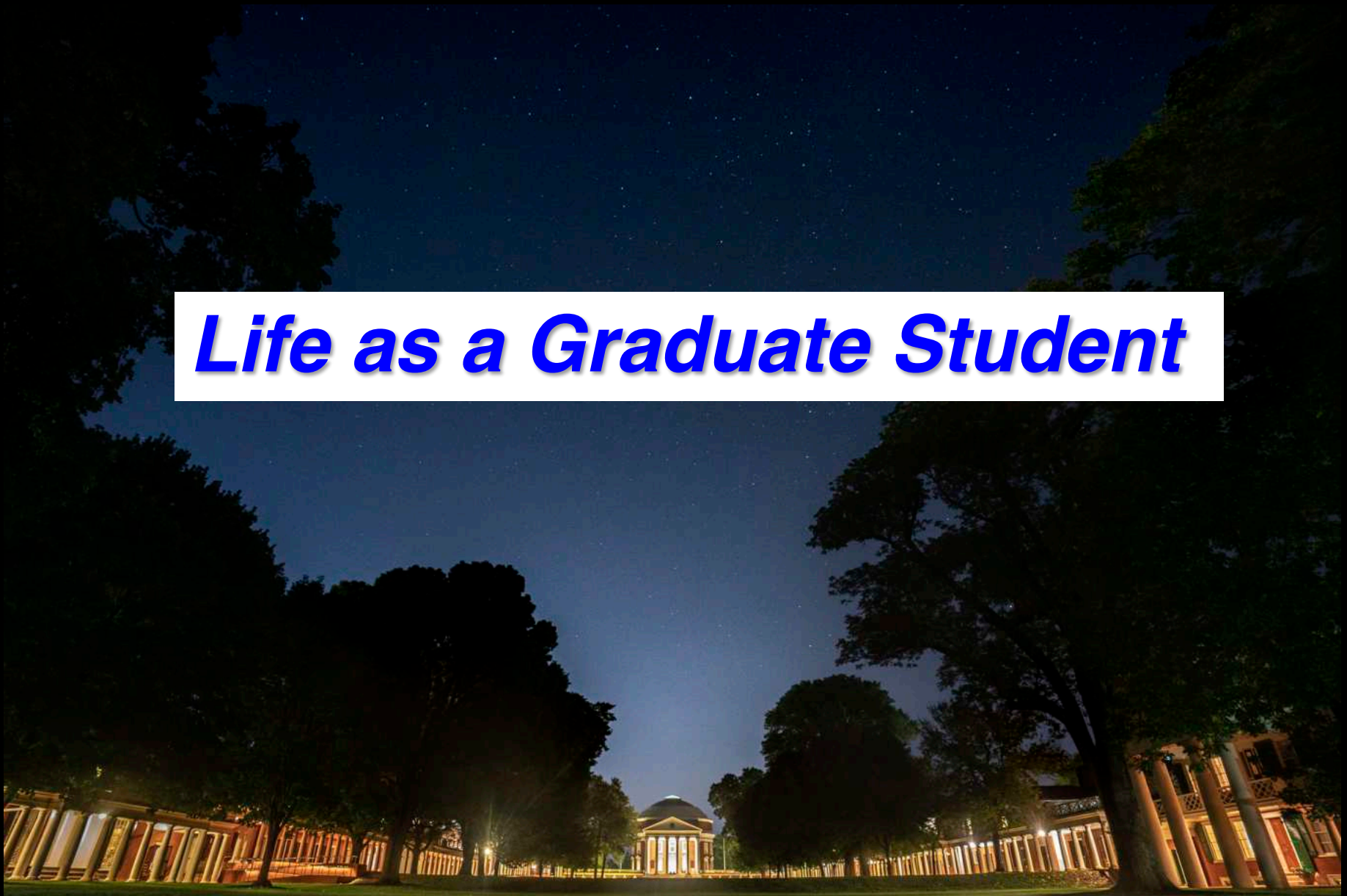
Spring 2024

Robert W. O'Connell

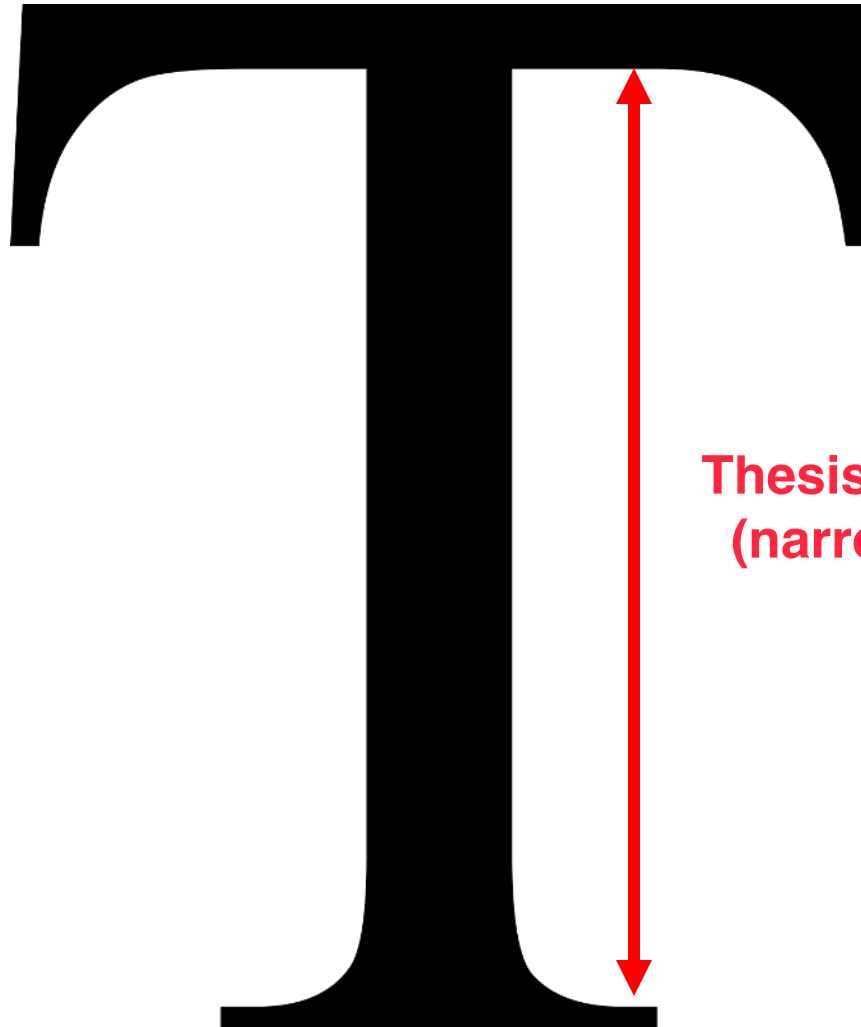
~95(?)%

95(?)% of astronomers
love *most* of what
they do

Life as a Graduate Student



Basics of many subfields (broad, shallow)
Coursework, colloquia, visitors, literature, etc



**Thesis research
(narrow, deep)**



Basics of many subfields (broad, shallow)
Coursework, colloquia, visitors, literature, etc

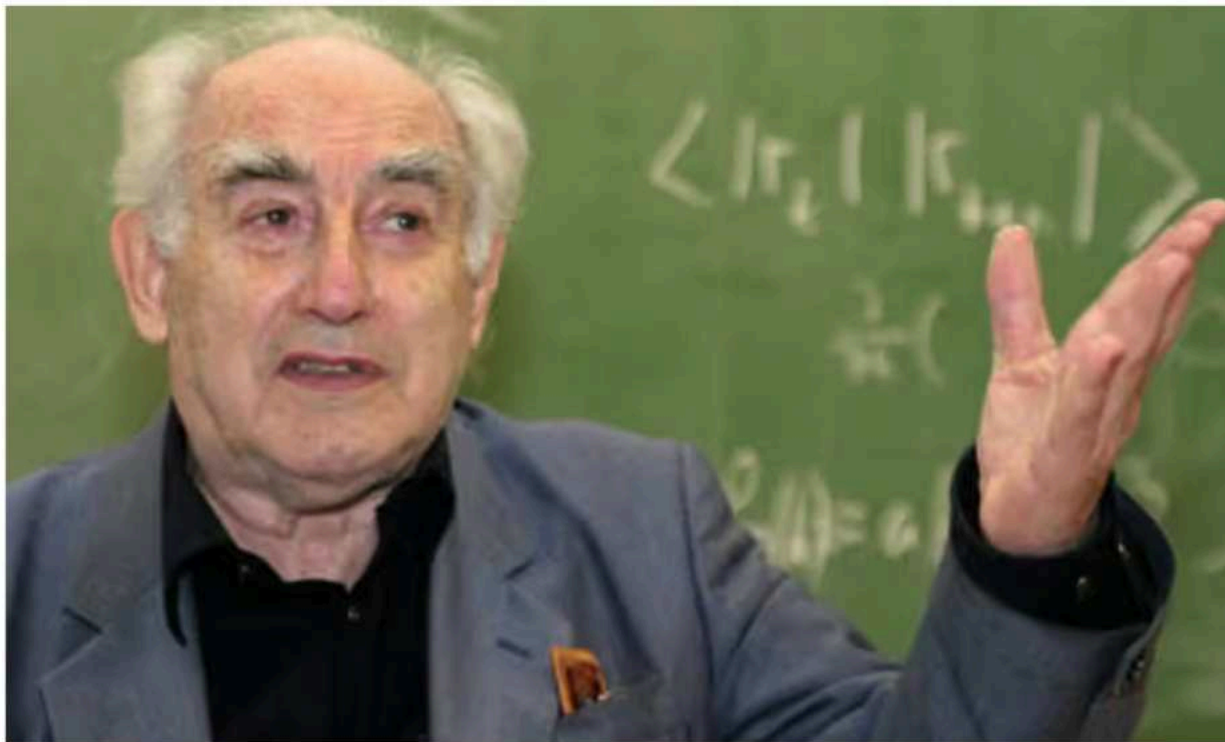


Thesis research
(narrow, deep)

**BOTH BROAD AND NARROW ASPECTS
IMPORTANT TO JOB PROSPECTS**

"The optimum strategy for a scientist is to know something about everything and everything about something."

-- Vitaly Ginzburg



📷 Vitaly Ginzburg in 2003, the year he jointly won the Nobel physics prize. Photograph: Tatyana Makeyeva/AFP/Getty Images

10,000

= Number of HOURS of close engagement with a speciality before a person is ready to make important contributions

May involve actual *reconfiguration* of neural circuits(!)

10,000

= Number of HCPs engaged with a specific person is ready to make important decisions

Cost of actual *reconfiguration* of neural circuitry

**FROM POP-PSYCHOLOGY
TAKE WITH GRAIN OF SALT**

10,000

= Number of HCPs with engagement
with a specific person is ready to
make important decisions

Clock starts running at the actual *reconfiguration* of neural
circuitry

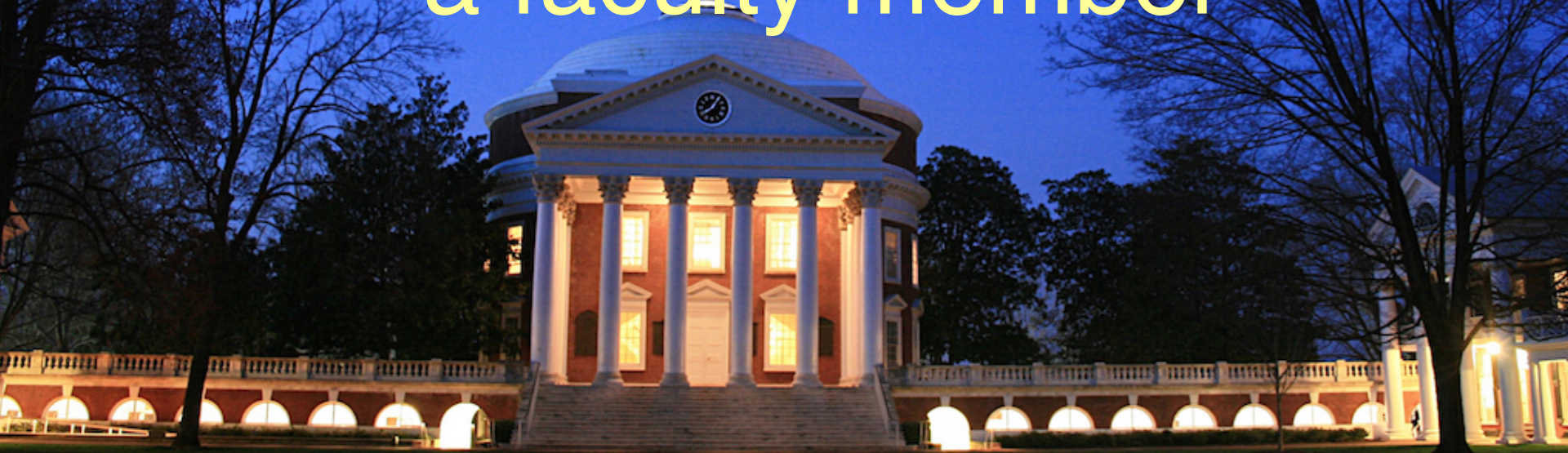
**CLOCK STARTS RUNNING
IN UNDERGRAD JUNIOR YEAR**

Life as a Faculty Member

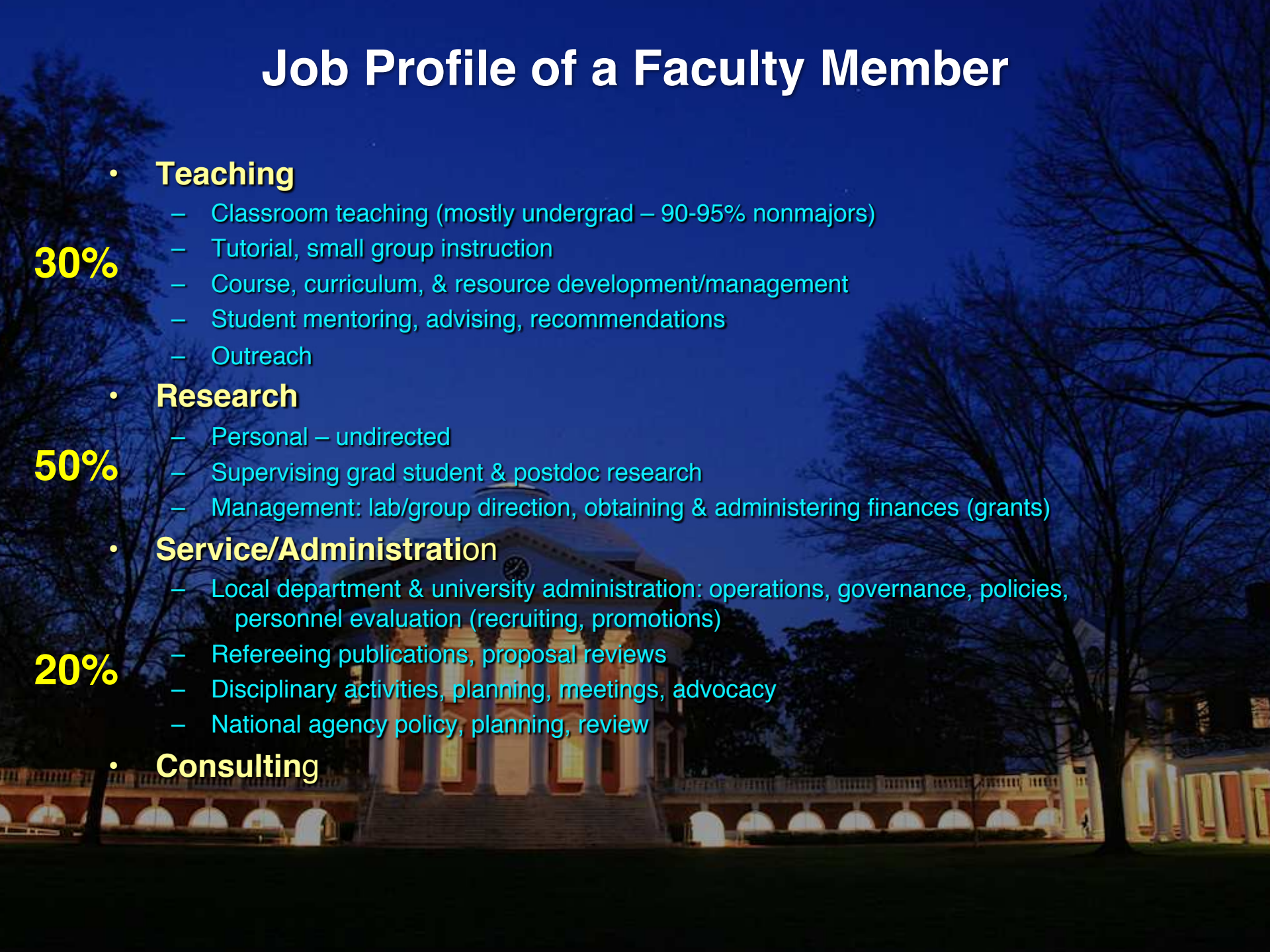


13

~ # Responsibilities of
a faculty member



Job Profile of a Faculty Member

- 
- **Teaching**
30%
 - Classroom teaching (mostly undergrad – 90-95% nonmajors)
 - Tutorial, small group instruction
 - Course, curriculum, & resource development/management
 - Student mentoring, advising, recommendations
 - Outreach
 - **Research**
50%
 - Personal – undirected
 - Supervising grad student & postdoc research
 - Management: lab/group direction, obtaining & administering finances (grants)
 - **Service/Administration**
20%
 - Local department & university administration: operations, governance, policies, personnel evaluation (recruiting, promotions)
 - Refereeing publications, proposal reviews
 - Disciplinary activities, planning, meetings, advocacy
 - National agency policy, planning, review
 - **Consulting**

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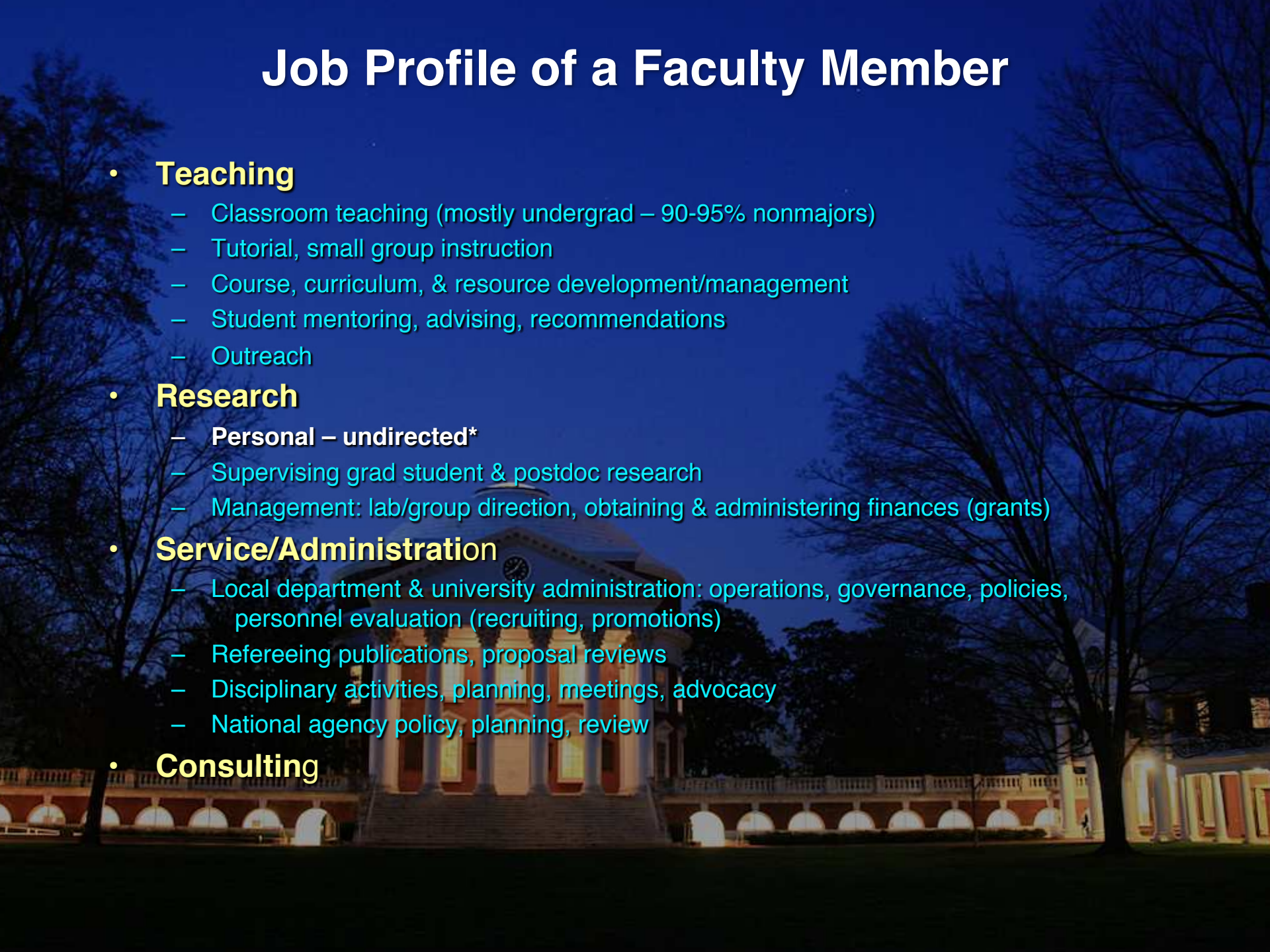
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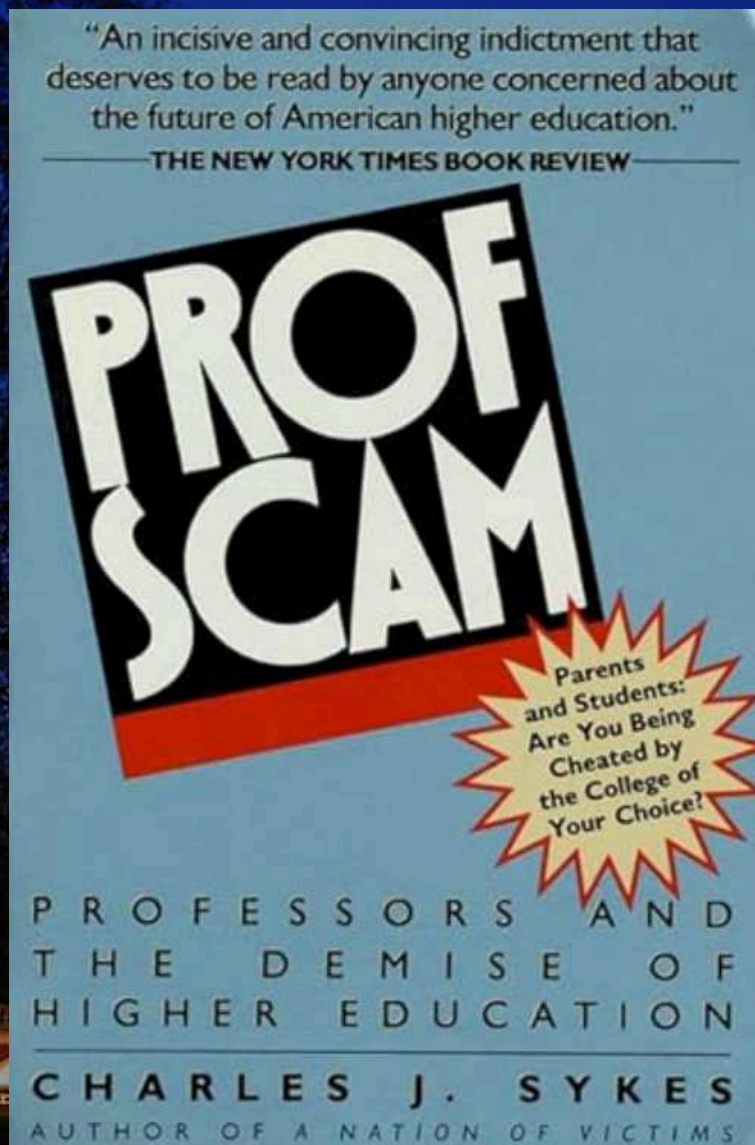
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Tenured profs replace
middle management



A Few Words About Tenure



A Few Words About Tenure

"An incisive and convincing indictment that deserves to be read by anyone concerned about the future of American higher education."

—THE NEW YORK TIMES BOOK REVIEW—

PROF

**Not just a transactional perk
A central organizing principle**

and Students
Are You Being
Cheated by
the College of
Your Choice?

PROFESSORS AND
THE DEMISE OF
HIGHER EDUCATION

CHARLES J. SYKES

AUTHOR OF A NATION OF VICTIMS

Tenure

**means never
having to work again**

imgflip.com

BENEFITS OF TENURE TO UNIVERSITIES

- o Tenure is the central organizing principle of academic program management at good universities.**
- o Ensures access to a large pool of talented candidate faculty**
- o 6-year vetting before tenure is awarded selects for people who are productive, committed, and self-motivated.**
- o Allows academic self-administration by tenured faculty. Very little middle management needed.**
- o The perk of tenure and absence of middle management reduce salaries and costs to students by at least 30%**
- o Drastically reduces faculty turnover costs**
- o Ensures a high quality, stable curriculum from leaders in their fields. Encourages innovation in teaching from experienced teachers.**
- o Ensures high quality research programs, often taking 5-10 years to develop. Almost by definition, no "Research-1" university can operate without tenure.**
- o Ensures good corporate memory and loyalty to the institution, critical for good administration**
- o Ensures a pool of expertise in a wide range of disciplines and specialties, essential for the common good and national security**

BENEFITS OF TENURE TO PROFESSORS

- o You have the presumption of continued employment as long as you discharge responsibilities effectively
- o Academic freedom: you can pursue research of your own choice; you can teach as you wish, within broad limits
- o You can be fired for cause, but you cannot be fired...
 - o if you're doing a good job...unless the university declares a fiscal emergency; you are insulated to a large extent from administrative incompetence.
 - o for administrative convenience
 - o for a personality or policy conflict with your superiors
 - o for what you think or write or say, in or out of class
- o You don't have to worry that specialization in research will make you unemployable in your mid- to late-career.

**Sadly, academic freedom has
been under assault from both
ends of the political spectrum**

The New York Times

The Fight Over Academic Freedom

Amid spiraling campus speech debates, many professors are rallying in defense of a bedrock principle. But can they agree on just what it means?

 Share full article



A gate at Harvard University showing the school's motto, "Veritas," or "Truth." Adam Glanzman for The New York Times



By Jennifer Schuessler

Feb. 16, 2024

BENEFITS OF TENURE TO PROFESSORS

- o You have the presumption of continued employment as long as you discharge responsibilities effectively
- o Academic freedom: you can pursue research of your choice; you can teach as you wish, with little interference
- o You can be fired for cause, but it's difficult to be fired...
 - o if you're doing good work, the university declares a fiscal emergency and cuts to a large extent from administrative salaries
 - o for reasons of convenience
 - o for policy conflict with your superiors
 - o if you think or write or say, in or out of class, something that makes you unemployable in your mid- to late-career.

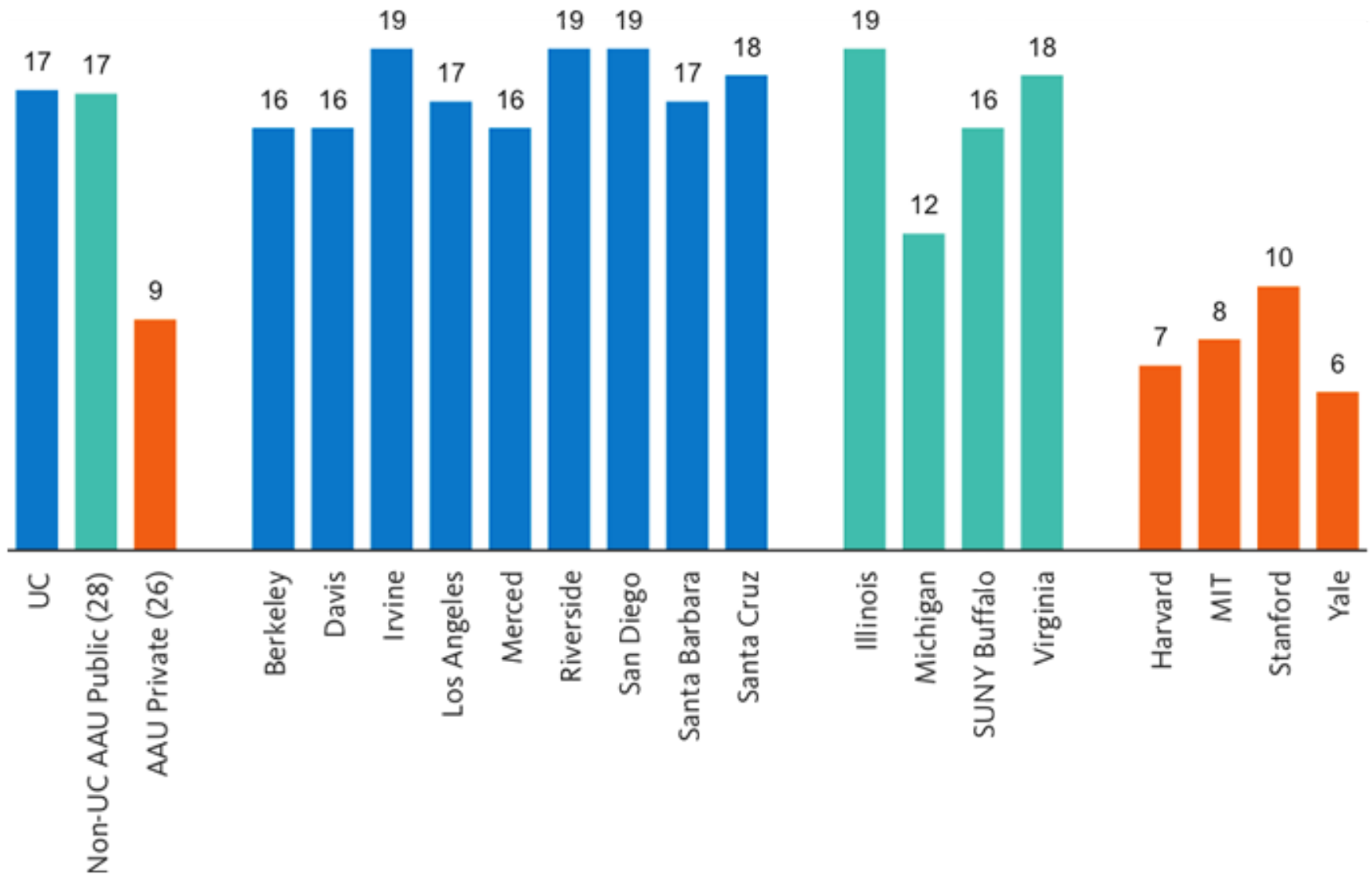
OPTIONAL TOPIC: "WHAT GOOD IS TENURE?"

~90-200

**= Number of UG students the
average professor must
teach each year**

= $10 \times S/F$ (in 3-credit classes)

Student/Faculty Ratios





...and here they are!

$\sim 7:1$

= Career averaged ratio of
total teaching time to
in-class time

Averaged effort for a 3-hour course: ~ 20 hours/wk

New burden for teachers!

Increasing emphasis on ELECTRONICS



*** One hour of course video takes 50-100 hours of prep**

1:1 to 3:1

= Ratio of real-time rehearsal
to delivery time for a well-
prepped talk

- An important class lecture
- A job talk
- A review talk
- A news conference
- etc



- An important class lecture
- A job talk
- A review talk
- A news conference
- etc



- An important class lecture
- A job talk
- A review talk
- A news conference
- etc



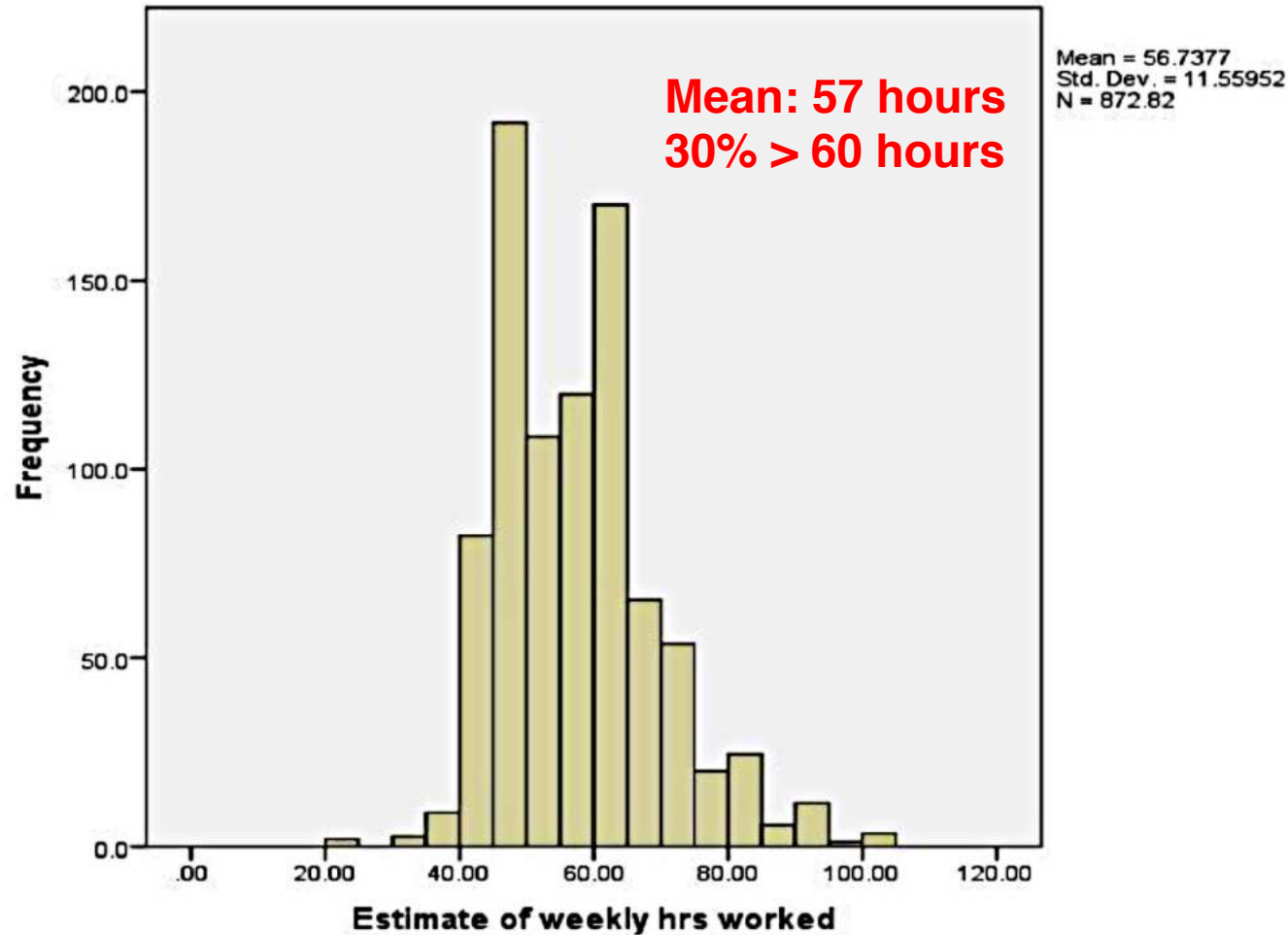
**MUST PLAN FOR HEAVY DEMANDS
OF TEACHING ON YOUR TIME**

~55

= Number of hours per
week professors
claim to work

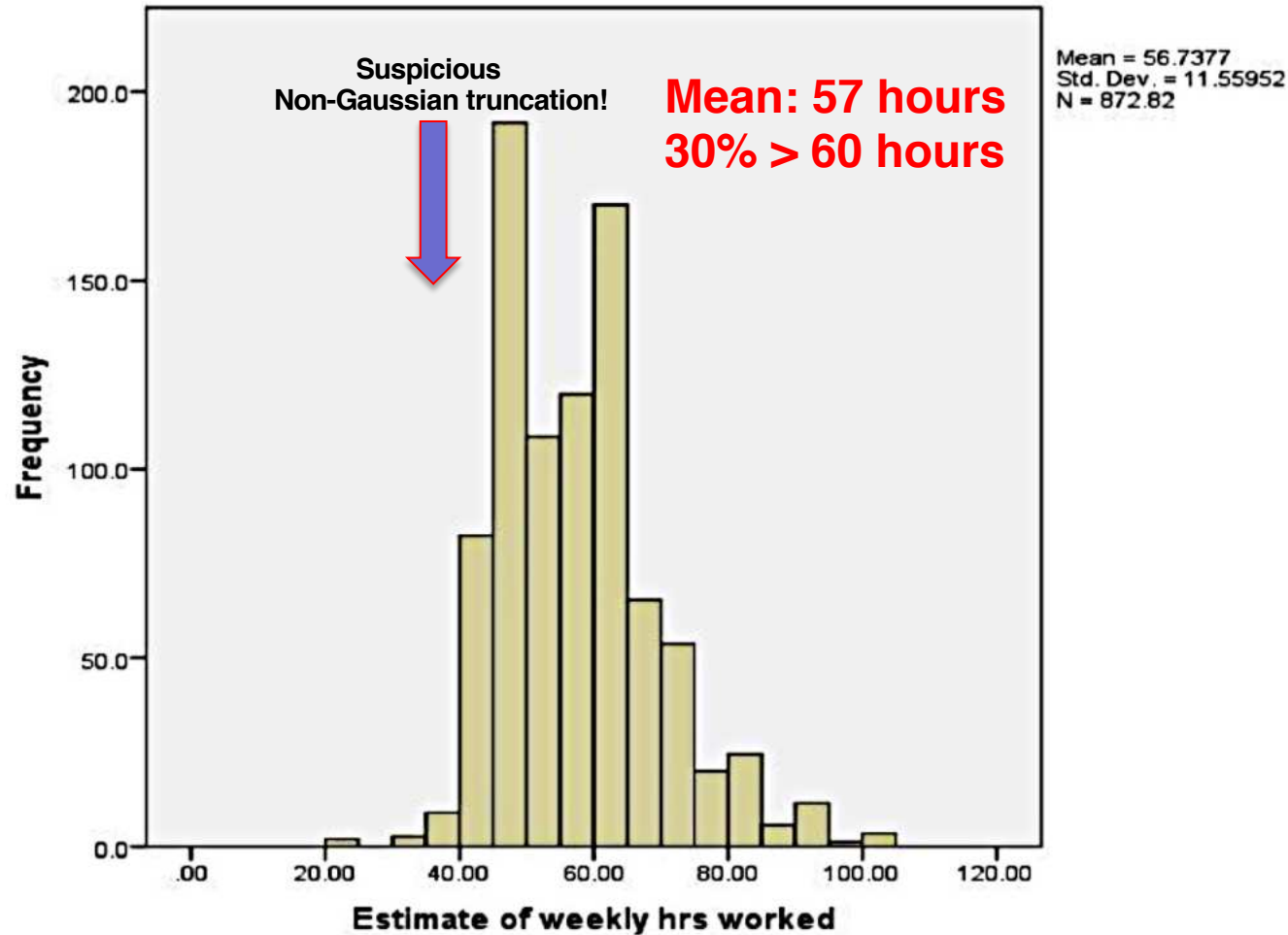
UVa Faculty Senate Survey (2012)

Figure VII-1: Frequency Distribution of Hours Worked Per Week, Full Time Faculty Only.



UVa Faculty Senate Survey (2012)

Figure VII-1: Frequency Distribution of Hours Worked Per Week, Full Time Faculty Only.



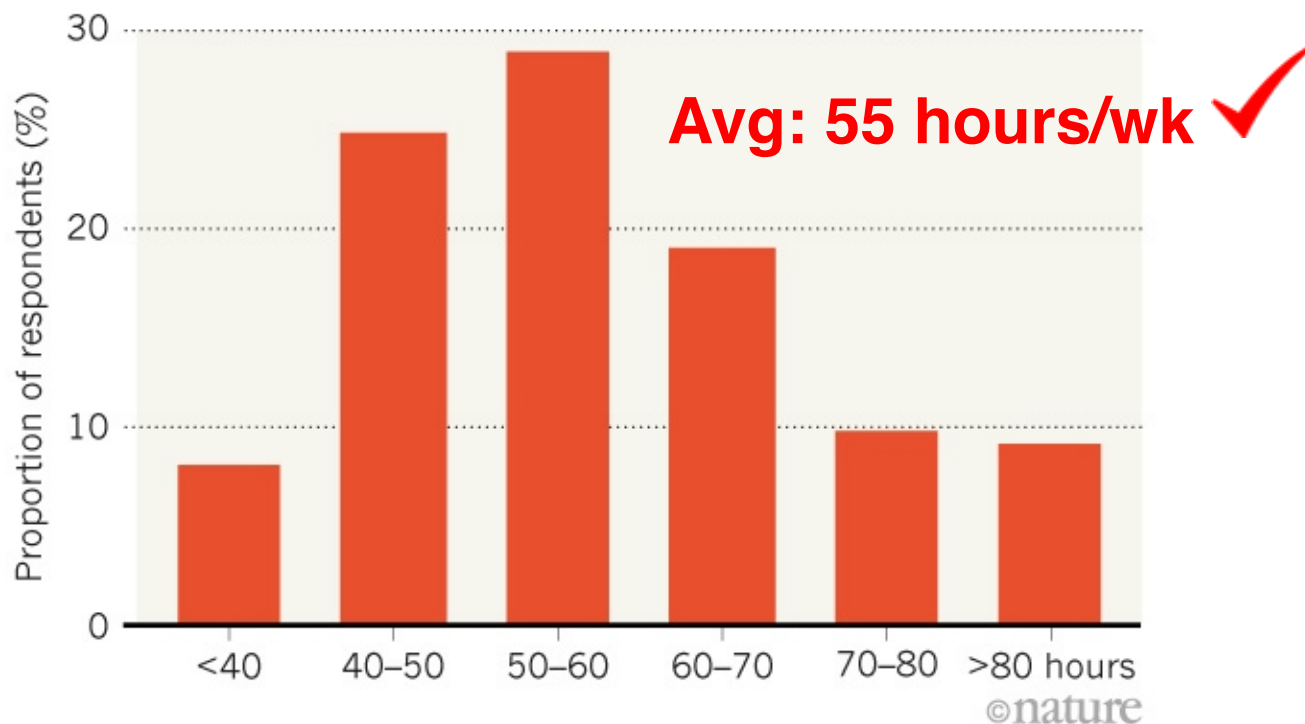
Scientist Survey: Hours Worked (Nature)

LONG HOURS

Some 38% of *Nature's* readers say they work more than 60 hours a week.

Poll question:

How many hours a week do you work on average? (12,869 responses)



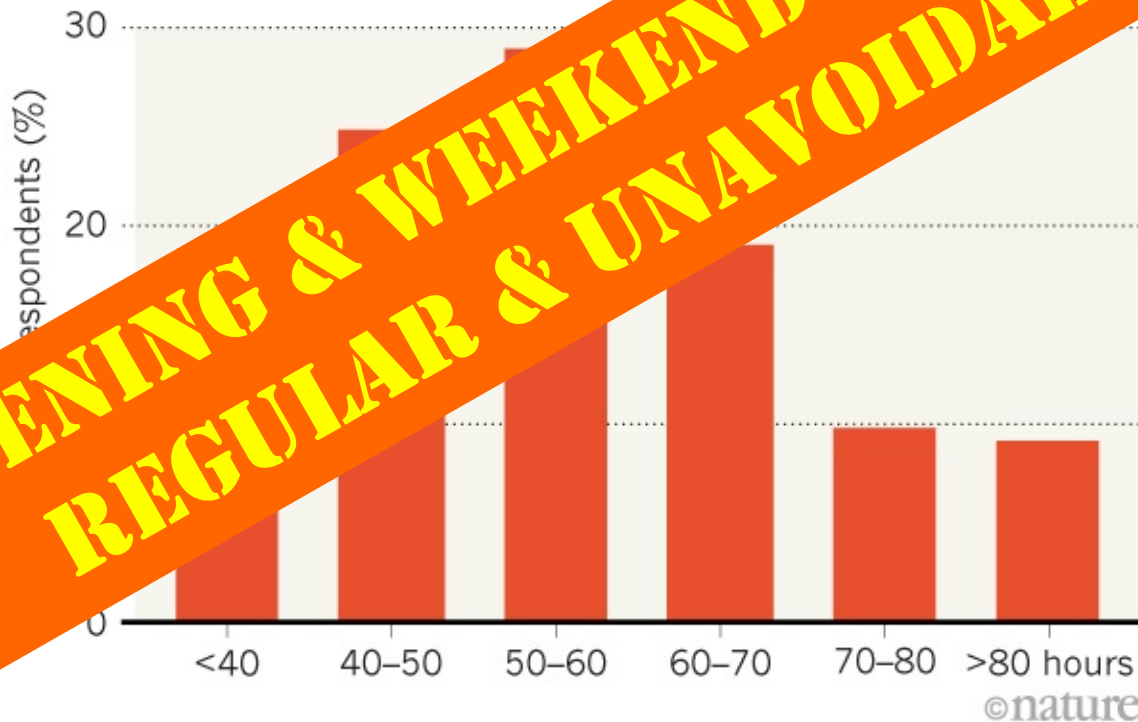
Scientist Survey: Hours Worked (Nature)

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Poll question:

How many hours a week do you work on



**EVENING & WEEKEND WORK IS
REGULAR & UNAVOIDABLE**

Work-Life Balance?

Can be good, but unlikely 9-5x5



Work-Life Balance?

Can be good, but unlikely 9-5x5



Most observing is done at night
and at remote sites.

Work-Life Balance?

Can be good, but unlikely 9-5x5

[Find Funding & Apply](#) ▾[Manage Your Award](#) ▾[Focus Areas](#) ▾[News & Events](#) ▾[About](#) ▾

Proposal & Award Policies & Procedures Guide (PAPPG)

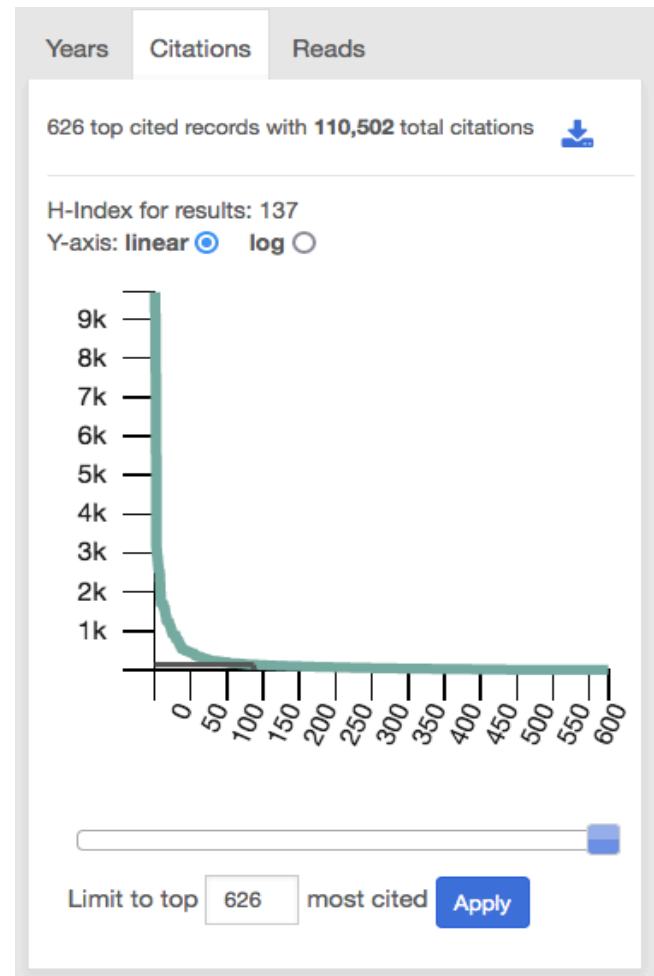
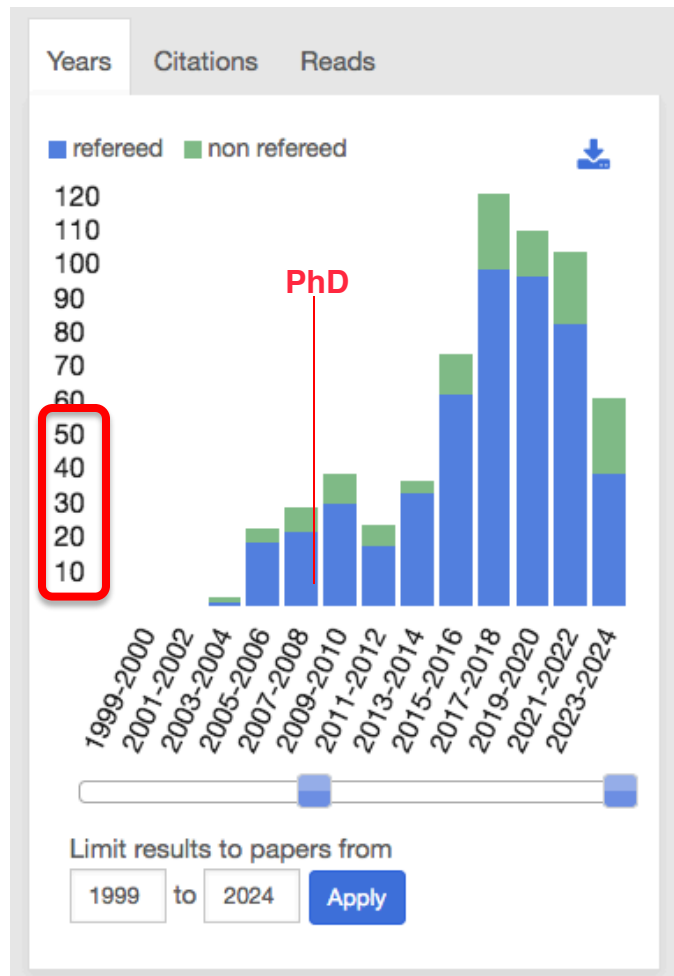
NSF 23-1: Effective for proposals submitted or due on or after January 30, 2023

[Home](#) / [NSF Proposal & Award Policies & Procedures Guide \(PAPPG\)](#) / [Proposal & Award Policies & Procedures Guide \(PAPPG\) \(NSF 23-1\)](#)

**Important deadlines are set by others
and are often inflexible.**

Work-Life Balance?

Can be good, but unlikely 9-5x5



Expected pace of research is set
by most productive outliers

Work-Life Balance?

Can be good, but unlikely 9-5x5



Astronomy is a global enterprise. Your colleagues can be in any of the 24 time zones.

Work-Life Balance?

Can be good, but unlikely 9-5x5



**NASA mission schedules constrained by budget,
technology development, person-power, orbital physics
-- not convenience of participants**

Work-Life Balance?

Can be good, but unlikely 9-5x5



You will have many opportunities to travel for “work,” often to nice places. Take advantage of this hidden perk.

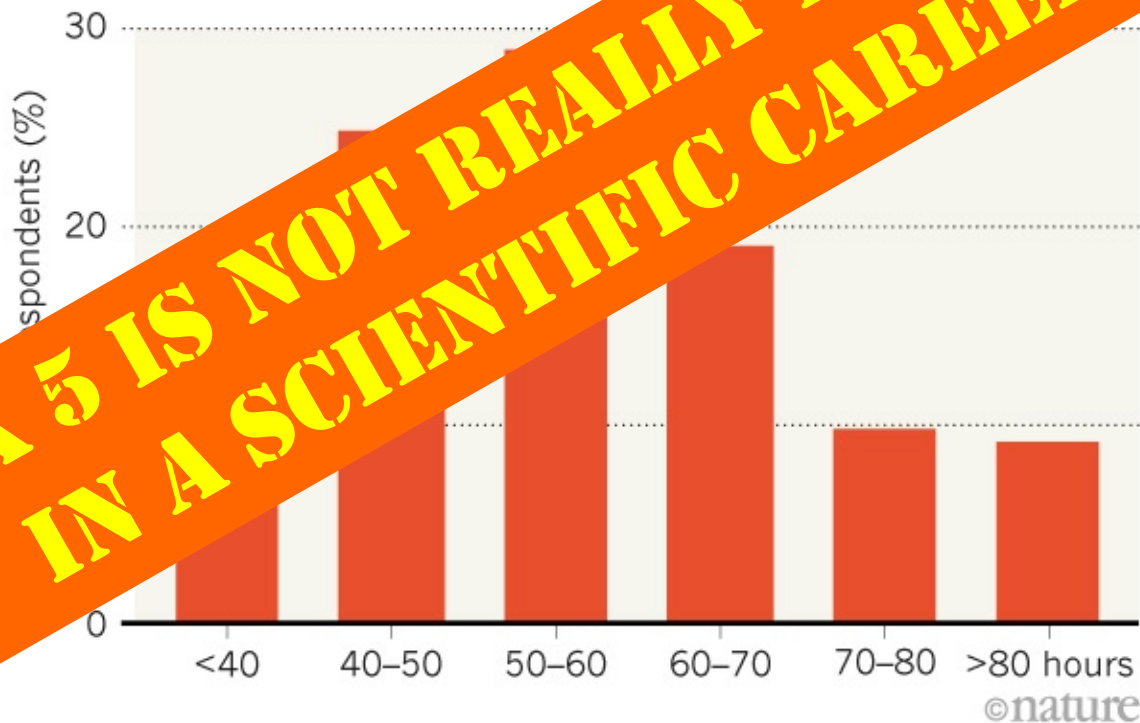
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How many hours a week do you work on



Scientist Survey: Hours Worked (Nature)

LONG HOURS

Some 38% of *Nature's* readers say they work more than 60 hours a week.

Poll question:

How many hours a week do you work on average? (Please select the closest answer)



Job Profile of a Faculty Member

- Teaching

- Classroom teaching (mostly undergrad)
- Tutorial, small group instruction
- Course, curriculum, & resource development/management
- Student mentoring, advising, recommendations
- Outreach

- Research

- Personal - undirected
- Supervising grad students
- Management: lab, administering finances (grants)

- Service/Admin

- Local administration: operations, governance, policies, personnel (hiring, promotions)
- Refereeing proposal reviews
- Disciplinary, planning, meetings, advocacy
- National agency policy, planning, review

- Consulting

**TIME MANAGEMENT IS
AN ESSENTIAL SKILL**

Job Profile of a Faculty Member

- Teaching

- Classroom teaching (mostly undergrad)
- Tutorial, small group instruction
- Course, curriculum, & resource development/management
- Student mentoring, advising, recommendations
- Outreach

- Research

- Personal -undirected
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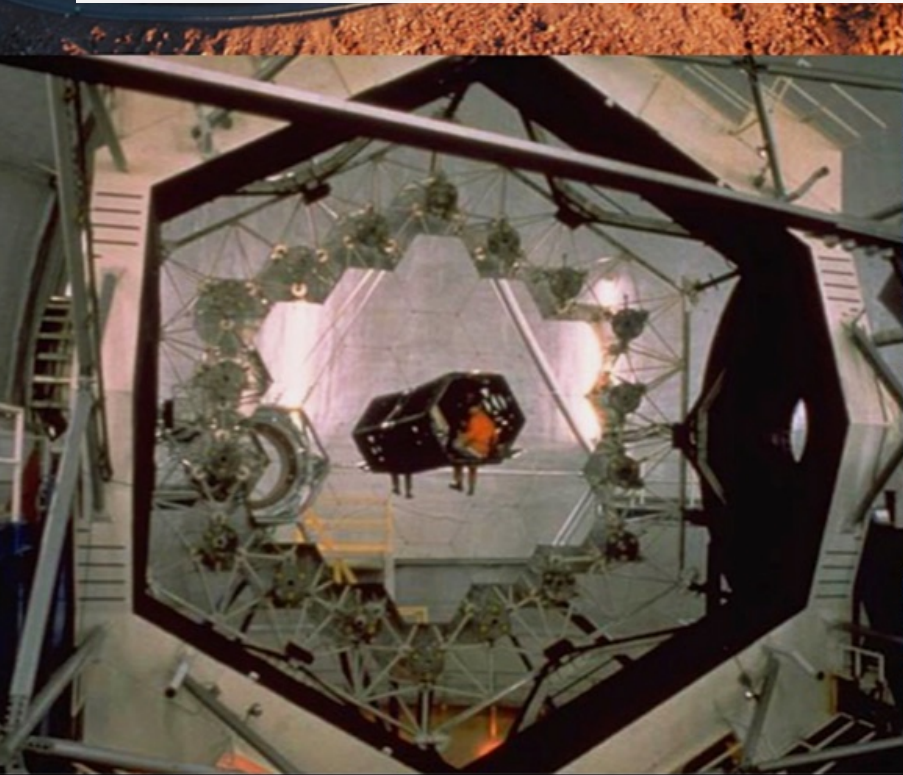
- Local administration: operations, governance, policies, personnel (hiring, promotions)
- Referee: proposal reviews
- Discipline: planning, meetings, advocacy
- National policy, planning, review

- Consulting

**OPTIONAL TOPIC:
"MANAGING YOUR TIME"**



Life as a Research Scientist



Job Profile of a Research Scientist

- **Research Support**
 - Observer support & training
 - Telescope time allocation
 - Software design, development, oversight
 - Data analysis pipelines, data archives, quality assurance
 - Instrumentation development
 - Documentation
 - Facility upgrade projects
 - Policy formulation
 - Personnel administration
- **Personal Research**
 - Allocation usually specified; typically 15-50% but wide variation
 - Grant support provides buy-outs of service time
- **General Service**
 - Refereeing publications, proposal reviews
 - Disciplinary activities, planning, meetings, advocacy
 - National agency policy, planning, review
- **Consulting**

Job Profile of a Research Scientist

- Research Support
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Job Profile of a Research Scientist

- Main occupational hazard of a research scientist?



Meetings

Astronomy Long-term PhD Employment Pattern Through the 2000's:

~1/3 Faculty

~1/3 Research Scientists

~1/3 Non Astronomy

} 65%

(Perley 2019)

"Non-Dedicated Astronomy"

Jobs drawing on general training in high-tech field

Examples:

Space science/applications (govt, contractors, commercial)

High-end computing (databases, AI)

Computational biology (genomics, neurology)

Communications (radio, microwave, fiber/laser)

Instrumentation (sensors, imaging, optics)

Medical imaging



Be alert for opportunities to develop transferable knowledge and skills

"Non-Dedicated Astronomy"

Jobs drawing on general training

Examples:

Space science/applications (e.g., commercial)

High-end computing

Computational biology (e.g., neurology)

Communications (e.g., microwave, fiber/laser)

Instrumentation (e.g., sensors, imaging, optics)

**OPTIONAL TOPIC: EMPLOYMENT
OUTSIDE ASTRONOMY**

**Be alert for opportunities to develop
transferable knowledge and skills**

3611

= Largest number of authors
on an astronomical paper

The Rise of Group Science

ADS Statistics on published Ast/Ap papers

# Authors	<u>1975</u>	<u>2016</u>
1	40%	7%
>2	26%	78%
>5	3%	39%
Max # to date	54	1187

3611

= Largest number of authors
on an astronomical paper.

Abbott et al., "Multi-Messenger Observations of a
Binary Neutron Star Merger," ApJL, 848, L12, 2017

Abbott et all !



Multi-messenger Observations of a Binary Neutron Star Merger*

LIGO Scientific Collaboration and Virgo Collaboration, Fermi GBM, INTEGRAL, IceCube Collaboration, AstroSat Cadmium Zinc Telluride Imager Team, IPN Collaboration, The Insight-HXMT Collaboration, ANTARES Collaboration, The Swift Collaboration, AGILE Team, The 1M2H Team, The Dark Energy Camera GW-EM Collaboration and the DES Collaboration, The DLT40 Collaboration, GRAVITA: GRAVitational Wave Inaf TeAm, The Fermi Large Area Telescope Collaboration, ATCA: Australia Telescope Compact Array, ASKAP: Australian SKA Pathfinder, Las Cumbres Observatory Group, OzGrav, DWF (Deeper, Wider, Faster Program), AST3, and CAASTRO Collaborations, The VINROUGE Collaboration, MASTER Collaboration, J-GEM, GROWTH, JAGWAR, Caltech-NRAO, TTU-NRAO, and NuSTAR Collaborations, Pan-STARRS, The MAXI Team, TZAC Consortium, KU Collaboration, Nordic Optical Telescope, ePESSTO, GROND, Texas Tech University, SALT Group, TOROS: Transient Robotic Observatory of the South Collaboration, The BOOTES Collaboration, MWA: Murchison Widefield Array, The CALET Collaboration, IKI-GW Follow-up Collaboration, H.E.S.S. Collaboration, LOFAR Collaboration, LWA: Long Wavelength Array, HAWC Collaboration, The Pierre Auger Collaboration, ALMA Collaboration, Euro VLBI Team, Pi of the Sky Collaboration, The Chandra Team at McGill University, DFN: Desert Fireball Network, ATLAS, High Time Resolution Universe Survey, RIMAS and RATIR, and SKA South Africa/MeerKAT (See the end matter for the full list of authors.)

Received 2017 October 3; revised 2017 October 6; accepted 2017 October 6; published 2017 October 16

Abstract

On 2017 August 17 a binary neutron star coalescence candidate (later designated GW170817) with merger time 12:41:04 UTC was observed through gravitational waves by the Advanced LIGO and Advanced Virgo detectors. The *Fermi* Gamma-ray Burst Monitor independently detected a gamma-ray burst (GRB 170817A) with a time delay of ~ 1.7 s with respect to the merger time. From the gravitational-wave signal, the source was initially localized to a sky region of 31 deg^2 at a luminosity distance of 40^{+8}_{-5} Mpc and with component masses consistent with neutron stars. The component masses were later measured to be in the range 0.86 to $2.26 M_{\odot}$. An extensive observing campaign was launched across the electromagnetic spectrum leading to the discovery of a bright optical transient (SSS17a, now with the IAU identification of AT 2017gfo) in NGC 4993 (at ~ 40 Mpc) less than 11 hours after the merger by the One-Meter, Two Hemisphere (1M2H) team using the 1 m Swope Telescope. The optical transient was independently detected by multiple teams within an hour. Subsequent observations targeted the object and its environment. Early ultraviolet observations revealed a blue transient that faded within 48 hours. Optical and infrared observations showed a redward evolution over ~ 10 days. Following early non-detections, X-ray and radio emission were discovered at the transient's position ~ 9 and ~ 16 days, respectively, after the merger. Both the X-ray and radio emission likely arise from a physical process that is distinct from the one that generates the UV/optical/near-infrared emission. No ultra-high-energy gamma-rays and no neutrino candidates consistent with the source were found in follow-up searches. These observations support the hypothesis that GW170817 was produced by the merger of two neutron stars in NGC 4993 followed by a short gamma-ray burst (GRB 170817A) and a kilonova/macronova powered by the radioactive decay of *r*-process nuclei synthesized in the ejecta.

Key words: gravitational waves – stars: neutron

1. Introduction

Over 80 years ago Baade & Zwicky (1934) proposed the idea of neutron stars, and soon after, Oppenheimer & Volkoff (1939) carried out the first calculations of neutron star models. Neutron stars entered the realm of observational astronomy in the 1960s by providing a physical interpretation of X-ray emission from Scorpius X-1 (Giacconi et al. 1962; Shklovsky 1967) and of radio pulsars (Gold 1968; Hewish et al. 1968; Gold 1969).

The discovery of a radio pulsar in a double neutron star system by Hulse & Taylor (1975) led to a renewed interest in binary stars and compact-object astrophysics, including the

development of a scenario for the formation of double neutron stars and the first population studies (Flannery & van den Heuvel 1975; Masetvitch et al. 1976; Clark 1979; Clark et al. 1979; Dewey & Cordes 1987; Lipunov et al. 1987; for reviews see Kalogera et al. 2007; Postnov & Yungelson 2014). The Hulse-Taylor pulsar provided the first firm evidence (Taylor & Weisberg 1982) of the existence of gravitational waves (Einstein 1916, 1918) and sparked a renaissance of observational tests of general relativity (Damour & Taylor 1991, 1992; Taylor et al. 1992; Wex 2014). Merging binary neutron stars (BNSs) were quickly recognized to be promising sources of detectable gravitational waves, making them a primary target for ground-based interferometric detectors (see Abadie et al. 2010 for an overview). This motivated the development of accurate models for the two-body, general-relativistic dynamics (Blanchet et al. 1995; Buonanno & Damour 1999; Pretorius 2005; Baker et al. 2006; Campanelli et al. 2006; Blanchet 2014) that are critical for detecting and interpreting gravitational waves (Abbott et al. 2016c, 2016d, 2016e, 2017a, 2017c, 2017d).

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GROUPS,
not people

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